# Glasgow Clustering

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# 1 How to use this document

This file is meant to be a **reproducible template** for a joint-board-area clustering study.

To run the same workflow for another joint board area, update:

- `params$target_board`
- `params$chosen_k_raw`
- `params$chosen_k_pc`
- `params$cmeans_ref_m_raw`

- `params$cmeans_ref_m_pc`
- `params$k_grid`
- `params$cmeans_m_grid`
- `params$cmeans_main_ambiguity_threshold`
- `params$cmeans_secondary_ambiguity_threshold`
- `params$output_dir`

The workflow is:

1. **Data preprocessing**
   - missingness
   - descriptive statistics
   - distributions
   - correlation structure
   - domain-wise PCA
2. **Clustering**
   - K-means (headline partitioning method)
   - Hierarchical clustering (hard-partition benchmark)
   - C-means (sensitivity / uncertainty layer)
   - Finite mixture models (secondary model-based comparator)
3. **Comparison**
   - cross-tables for selected clustering results
   - combined diagnostic plots
   - cluster maps
   - difference maps

The design follows the same analysis logic already used in your exploratory document: missingness review, descriptive statistics, domain-grouped distributions, correlation heatmaps, and domain-wise PCA before clustering.

# 2 Shared objects

# 3 Helper functions

# 4 Data preprocessing

## 4.1 Read and filter the data

## 4.2 Missingness

Table 1: Indicators with missing values in Glasgow

| Indicator | Missing_Count | Percent_Missing |
|---|---|---|
| Attendance | 86 | 11.53 |
| Attainment | 46 | 6.17 |
| crime_rate | 8 | 1.07 |
| crime_count | 7 | 0.94 |
| CIF | 3 | 0.40 |
| not_participating | 3 | 0.40 |
| ALCOHOL | 2 | 0.27 |
| DRUG | 2 | 0.27 |
| EMERG | 2 | 0.27 |
| SMR | 2 | 0.27 |

| | | |
|---|---|---|
| University | 2 | 0.27 |
| broadband | 2 | 0.27 |
| DEPRESS | 1 | 0.13 |
| LBWT | 1 | 0.13 |

```
## Indicators retained after missingness filtering: 31
```

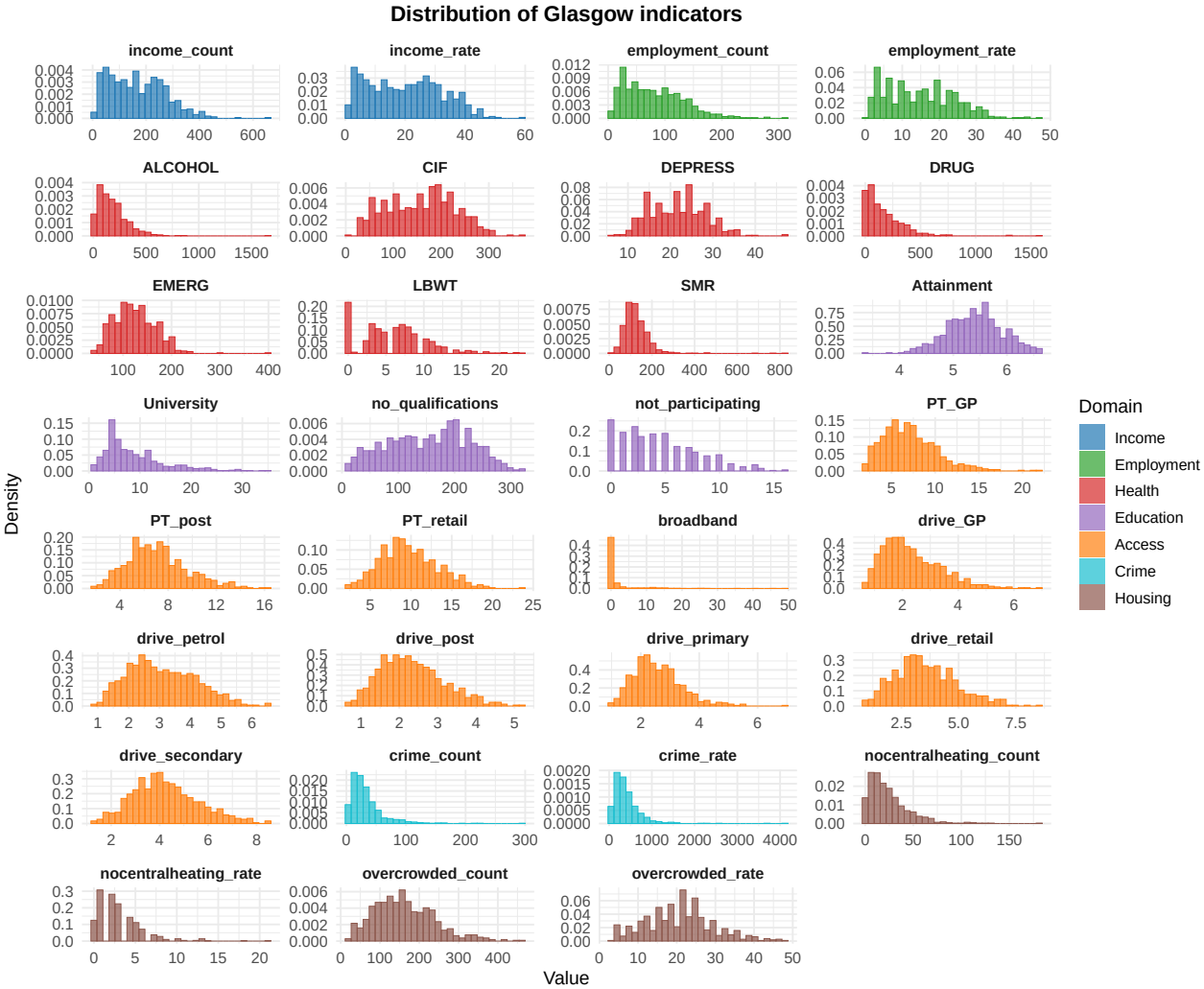
```
## Indicators dropped: Attendance
```

### 4.3 Statistics

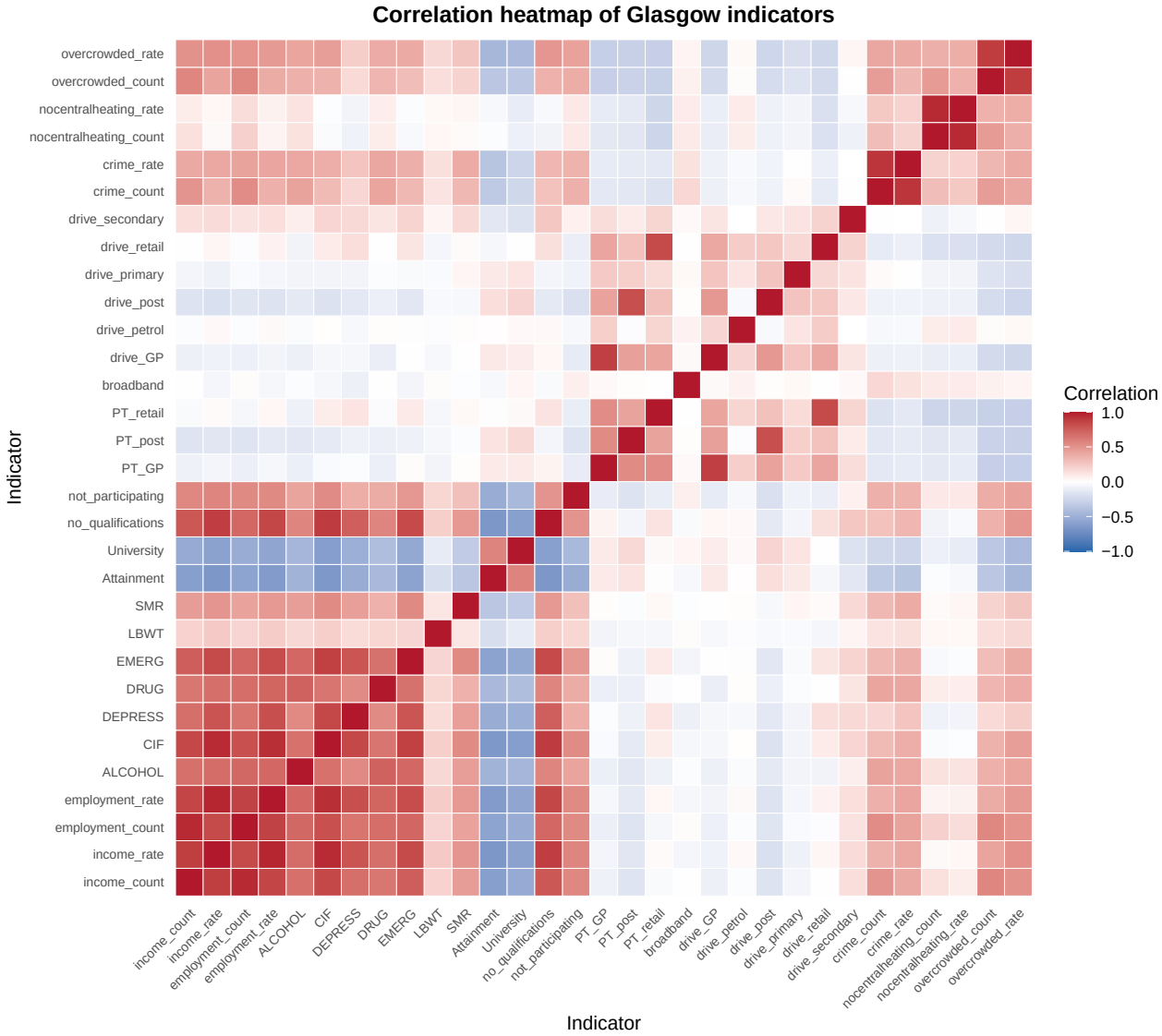
Table 2: Descriptive statistics for Glasgow analysis data

| Indicator | n | mean | median | sd | skewness | kurtosis | Normality_Test | p_value |
|---|---|---|---|---|---|---|---|---|
| Attainment | 696 | 5.40 | 5.41 | 0.52 | -0.12 | -0.06 | Shapiro-Wilk | 0.0534 |
| crime_rate | 696 | 407.41 | 326.00 | 352.66 | 4.23 | 32.05 | Shapiro-Wilk | <0.0001 |
| crime_count | 696 | 33.79 | 26.00 | 29.72 | 3.07 | 15.59 | Shapiro-Wilk | <0.0001 |
| CIF | 696 | 157.39 | 160.00 | 70.86 | 0.02 | -0.86 | Shapiro-Wilk | <0.0001 |
| not_participating | 696 | 4.40 | 4.00 | 3.54 | 0.72 | -0.12 | Shapiro-Wilk | <0.0001 |
| ALCOHOL | 696 | 179.16 | 152.00 | 142.59 | 2.55 | 17.39 | Shapiro-Wilk | <0.0001 |
| DRUG | 696 | 161.07 | 117.50 | 168.31 | 2.57 | 13.19 | Shapiro-Wilk | <0.0001 |
| EMERG | 696 | 124.75 | 122.00 | 42.31 | 0.70 | 2.09 | Shapiro-Wilk | <0.0001 |
| SMR | 696 | 126.13 | 116.00 | 67.14 | 4.38 | 36.08 | Shapiro-Wilk | <0.0001 |
| University | 696 | 8.68 | 7.00 | 5.87 | 1.40 | 2.14 | Shapiro-Wilk | <0.0001 |
| broadband | 696 | 1.43 | 0.00 | 4.80 | 5.40 | 37.40 | Shapiro-Wilk | <0.0001 |
| DEPRESS | 696 | 21.74 | 22.00 | 6.38 | 0.22 | -0.15 | Shapiro-Wilk | <0.0001 |
| LBWT | 696 | 5.93 | 6.00 | 4.38 | 0.69 | 0.65 | Shapiro-Wilk | <0.0001 |
| PT_GP | 696 | 7.09 | 6.68 | 3.10 | 0.98 | 1.56 | Shapiro-Wilk | <0.0001 |
| PT_post | 696 | 7.08 | 6.83 | 2.50 | 0.63 | 0.33 | Shapiro-Wilk | <0.0001 |
| PT_retail | 696 | 9.72 | 9.32 | 3.46 | 0.44 | 0.03 | Shapiro-Wilk | <0.0001 |
| drive_GP | 696 | 2.32 | 2.11 | 1.04 | 0.97 | 1.02 | Shapiro-Wilk | <0.0001 |
| drive_petrol | 696 | 3.09 | 2.95 | 1.10 | 0.40 | -0.42 | Shapiro-Wilk | <0.0001 |
| drive_post | 696 | 2.33 | 2.24 | 0.83 | 0.54 | -0.03 | Shapiro-Wilk | <0.0001 |
| drive_primary | 696 | 2.57 | 2.45 | 0.83 | 0.90 | 1.54 | Shapiro-Wilk | <0.0001 |
| drive_retail | 696 | 3.56 | 3.40 | 1.34 | 0.52 | 0.07 | Shapiro-Wilk | <0.0001 |
| drive_secondary | 696 | 4.24 | 4.07 | 1.35 | 0.46 | -0.03 | Shapiro-Wilk | <0.0001 |
| employment_count | 696 | 81.01 | 72.50 | 52.02 | 0.84 | 0.69 | Shapiro-Wilk | <0.0001 |
| employment_rate | 696 | 14.82 | 14.00 | 9.07 | 0.43 | -0.53 | Shapiro-Wilk | <0.0001 |
| income_count | 696 | 167.71 | 157.00 | 103.78 | 0.57 | 0.10 | Shapiro-Wilk | <0.0001 |
| income_rate | 696 | 20.65 | 21.00 | 12.10 | 0.22 | -0.91 | Shapiro-Wilk | <0.0001 |
| no_qualifications | 696 | 154.05 | 158.00 | 71.01 | -0.10 | -0.92 | Shapiro-Wilk | <0.0001 |
| nocentralheating_count | 696 | 24.12 | 18.00 | 21.46 | 2.03 | 7.00 | Shapiro-Wilk | <0.0001 |
| nocentralheating_rate | 696 | 3.03 | 2.00 | 2.62 | 1.95 | 6.38 | Shapiro-Wilk | <0.0001 |
| overcrowded_count | 696 | 162.35 | 153.50 | 76.73 | 0.66 | 0.48 | Shapiro-Wilk | <0.0001 |
| overcrowded_rate | 696 | 20.76 | 21.00 | 8.65 | 0.33 | 0.05 | Shapiro-Wilk | <0.0001 |

## 4.4 Distribution



## 4.5 Correlation




## 4.6 PCA

The two clustering feature sets keep the rate variables for Income, Employment, Crime, and Housing in both designs.

For the remaining domains:

- the **raw-indicator-based** set keeps all raw indicators with missingness below the threshold;
- the **PC-based** set replaces the remaining domains with domain-wise PCs whose **individual variance explained is greater than 10%**.

Table 3: Domain-wise comparison of variables retained and PC-based feature sets

| Domain | Retained raw indicators |
|---|---|
| Income | income_rate |
| Employment | employment_rate |
| Health | CIF, ALCOHOL, DRUG, SMR, DEPRESS, LBWT, EMERG |
| Education | Attainment, no_qualifications, not_participating, University |
| Access | drive_petrol, drive_GP, drive_post, drive_primary, drive_retail, drive_secondary, PT_GP, PT_post, PT |
| Crime | crime_rate |
| Housing | overcrowded_rate, nocentralheating_rate |

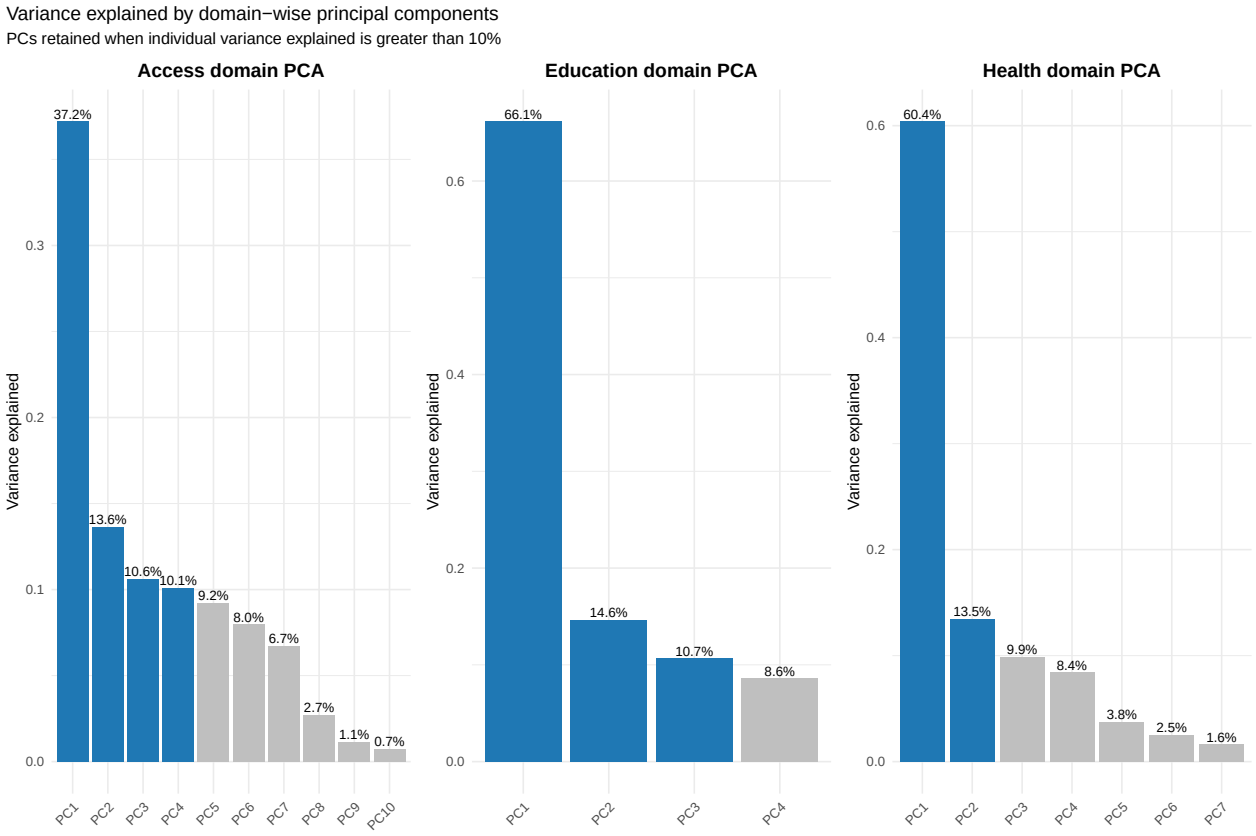



Table 4: PCA loadings for retained Access components

| Indicator | Access_PC1 | Access_PC2 | Access_PC3 | Access_PC4 |
|---|---|---|---|---|
| drive_petrol | 0.128 | 0.492 | -0.452 | -0.249 |
| drive_GP | 0.417 | -0.040 | -0.203 | -0.181 |
| drive_post | 0.358 | -0.491 | 0.020 | 0.030 |
| drive_primary | 0.209 | -0.078 | -0.285 | 0.244 |
| drive_retail | 0.358 | 0.429 | 0.281 | 0.032 |
| drive_secondary | 0.137 | 0.139 | 0.283 | 0.743 |
| PT_GP | 0.438 | 0.000 | -0.155 | -0.169 |
| PT_post | 0.382 | -0.429 | 0.066 | -0.016 |
| PT_retail | 0.391 | 0.348 | 0.282 | 0.008 |

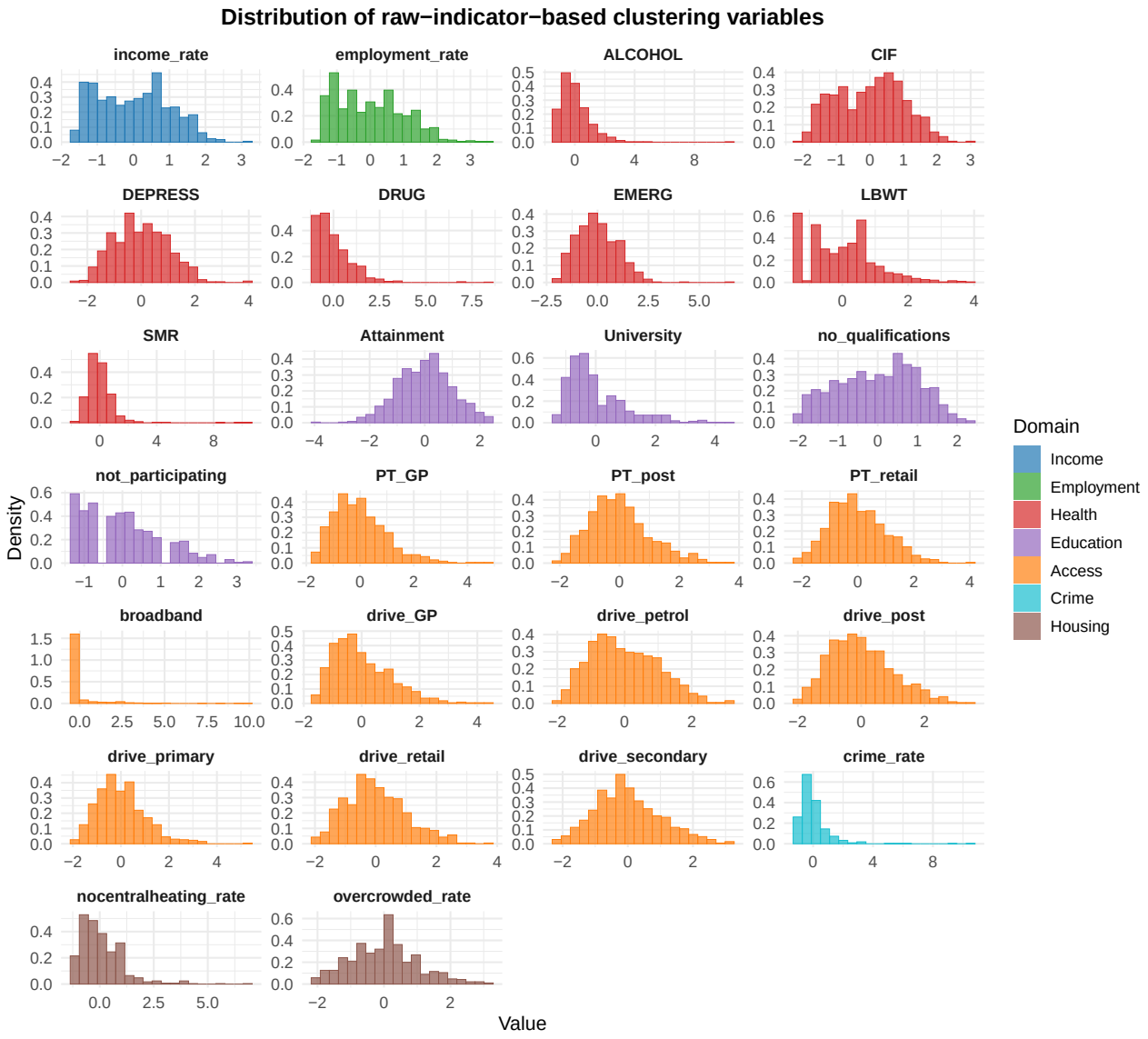
| | | | | |
|---|---|---|---|---|
| broadband | 0.019 | 0.043 | -0.638 | 0.513 |

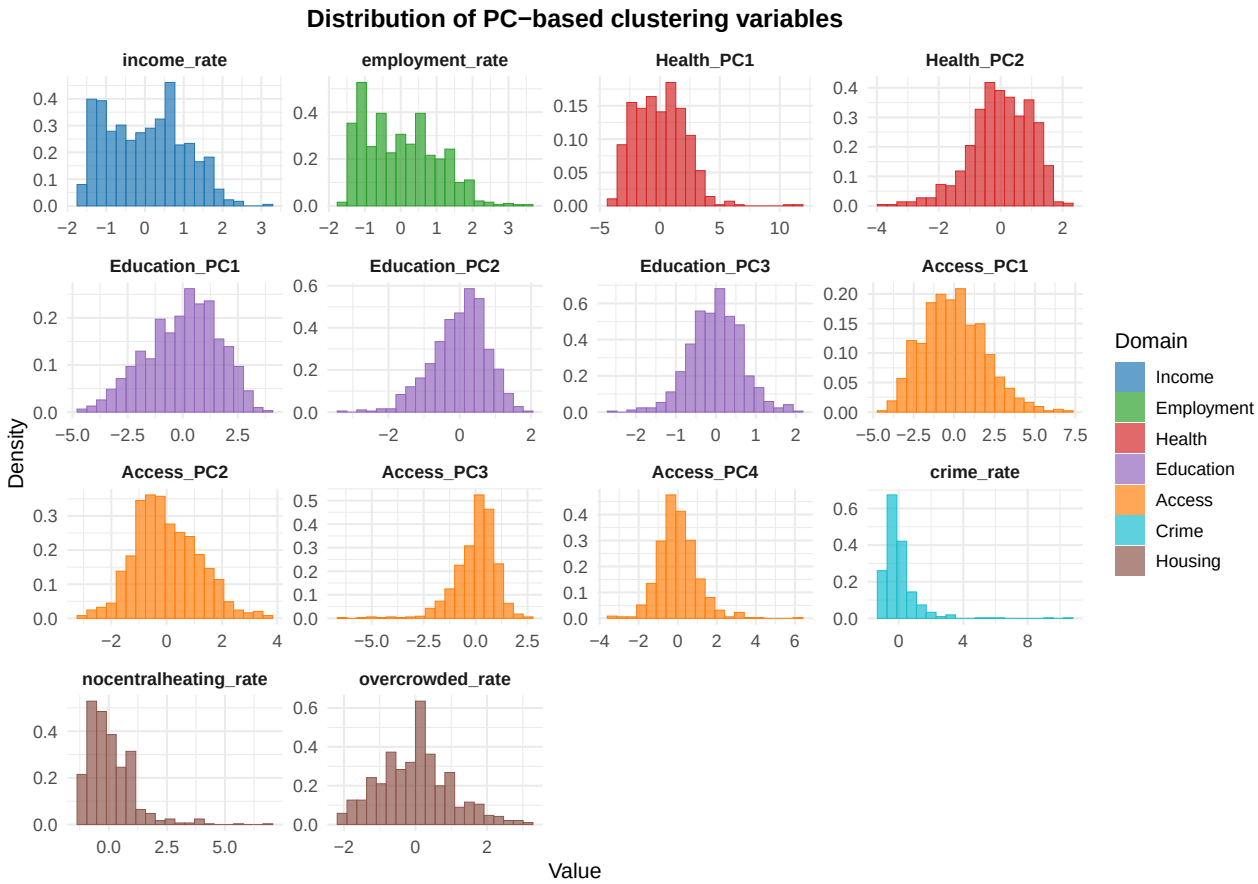
Table 5: PCA loadings for retained Education components

| Indicator | Education_PC1 | Education_PC2 | Education_PC3 |
|---|---|---|---|
| Attainment | -0.524 | -0.094 | -0.526 |
| no_qualifications | 0.525 | 0.201 | 0.372 |
| not_participating | 0.454 | -0.853 | -0.254 |
| University | -0.493 | -0.472 | 0.722 |

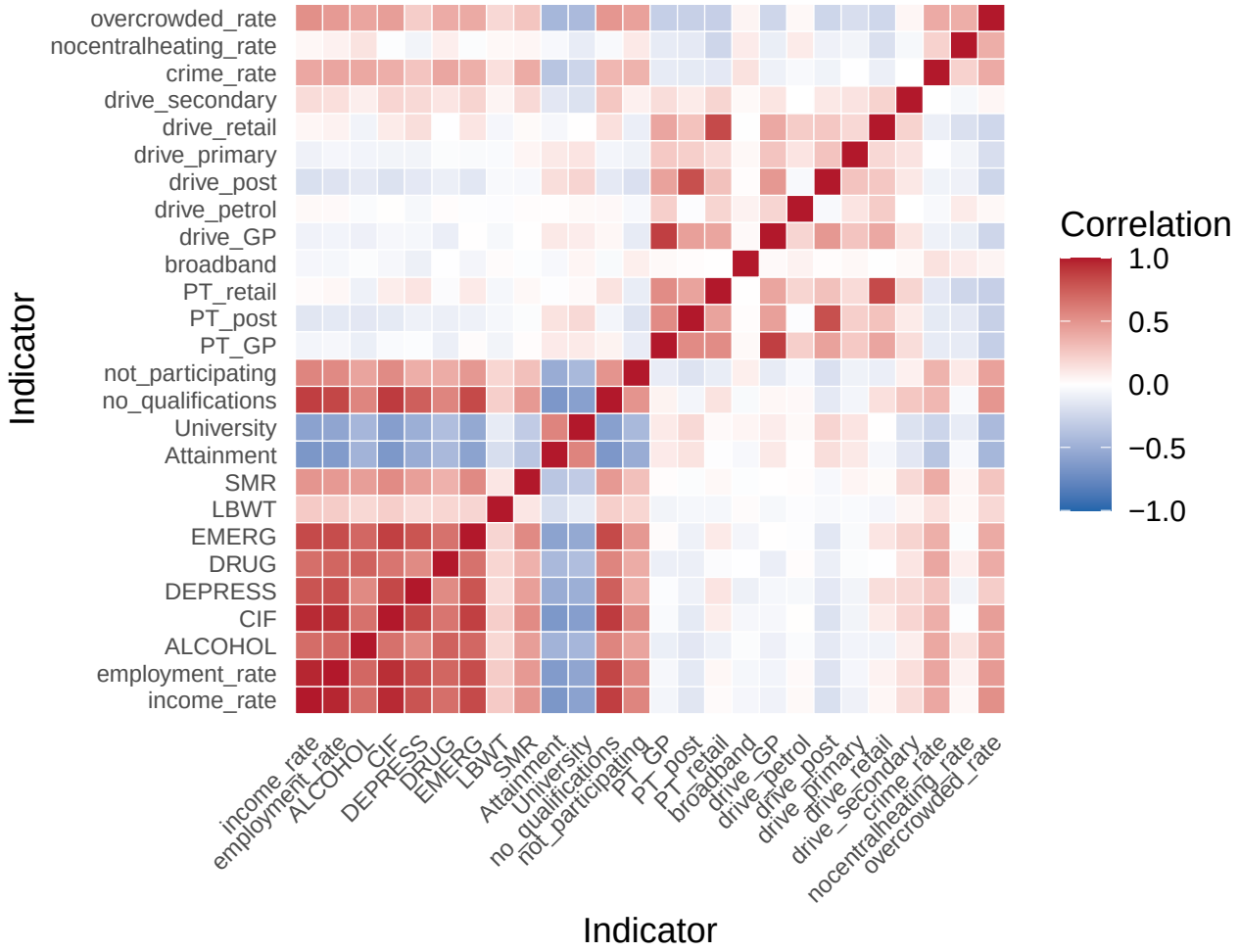
Table 6: PCA loadings for retained Health components

| Indicator | Health_PC1 | Health_PC2 |
|---|---|---|
| CIF | 0.451 | 0.039 |
| ALCOHOL | 0.400 | 0.026 |
| DRUG | 0.385 | -0.014 |
| SMR | 0.313 | 0.137 |
| DEPRESS | 0.411 | 0.086 |
| LBWT | 0.137 | -0.983 |
| EMERG | 0.450 | 0.074 |

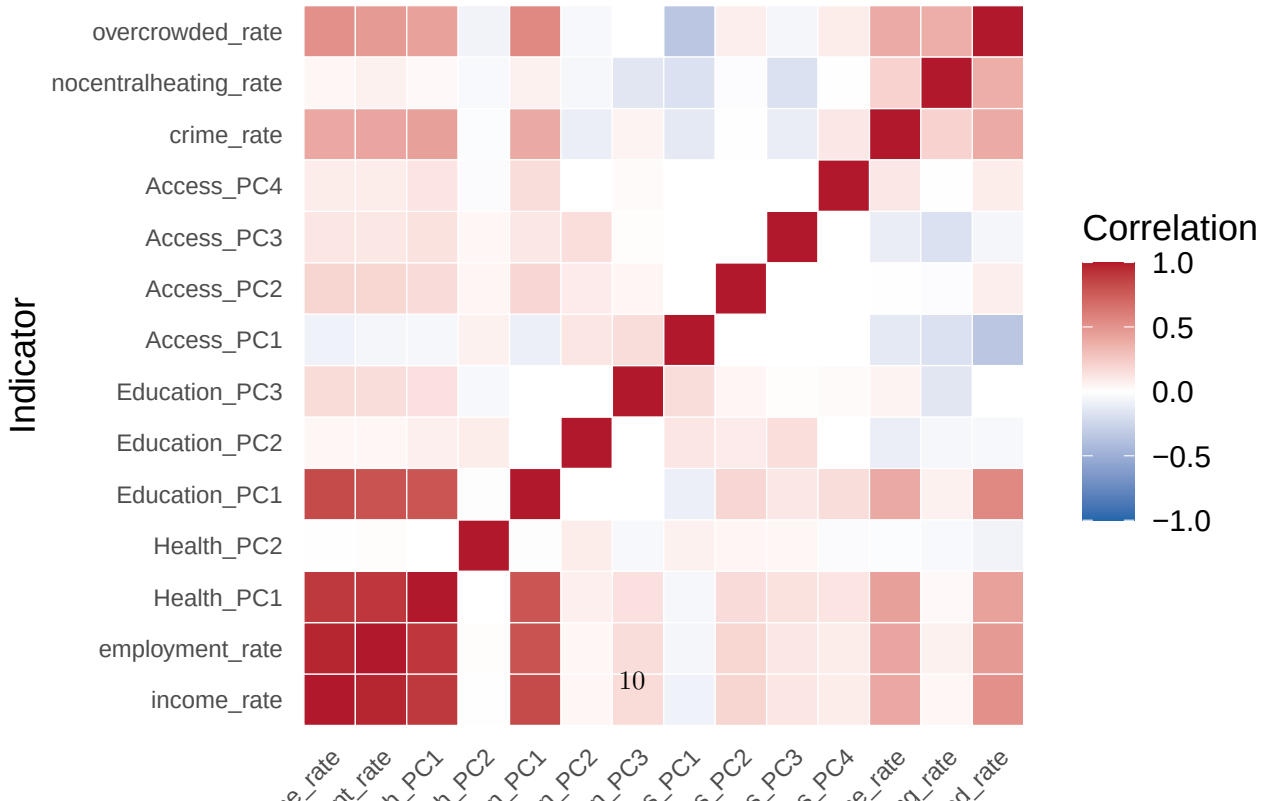

Distribution of raw−indicator−based clustering variables
income_rate
employment_rate
ALCOHOL
CIF
DEPRESS
DRUG
EMERG
LBWT
SMR
Attainment
University
no_qualifications
not_participating
PT_GP
PT_post
PT_retail
broadband
drive_GP
drive_petrol
drive_post
drive_primary
drive_retail
drive_secondary
crime_rate
nocentralheating_rate
overcrowded_rate
Density
Value
Domain
Income
Employment
Health
Education
Access
Crime
Housing


Distribution of PC−based clustering variables
income_rate
employment_rate
Health_PC1
Health_PC2
Education_PC1
Education_PC2
Education_PC3
Access_PC1
Access_PC2
Access_PC3
Access_PC4
crime_rate
nocentralheating_rate
overcrowded_rate
Density
Value
Domain
Income
Employment
Health
Education
Access
Crime
Housing

## Correlation heatmap: raw−indicator−based feature set




## Correlation heatmap: PC−based feature set



# 5 Clustering

## 5.1 K-means

### 5.1.1 Raw-based clustering

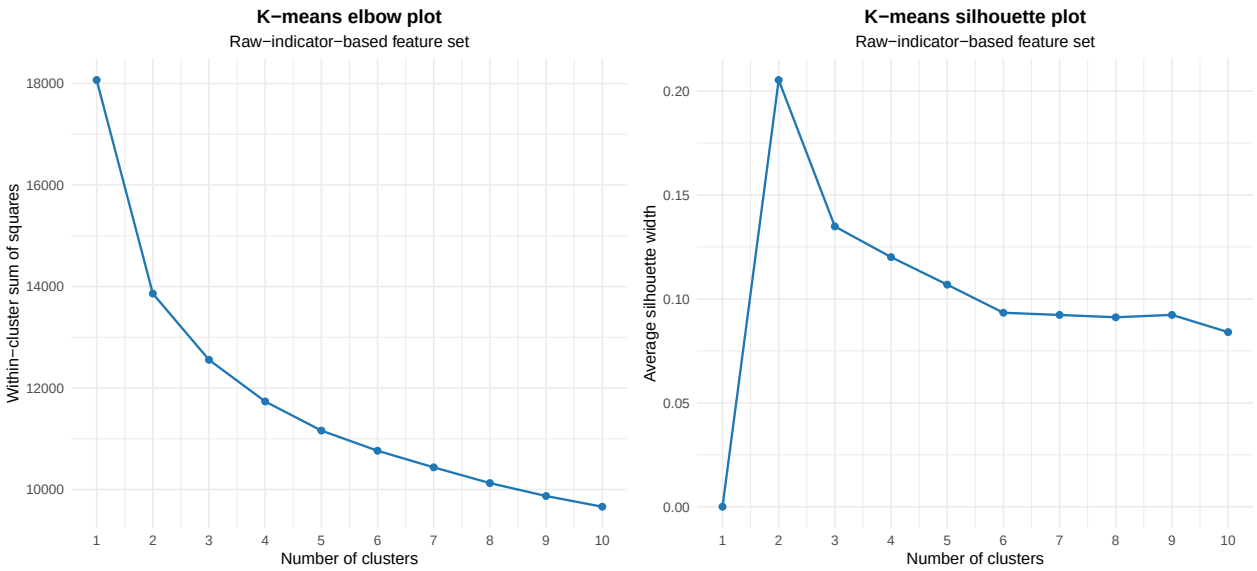
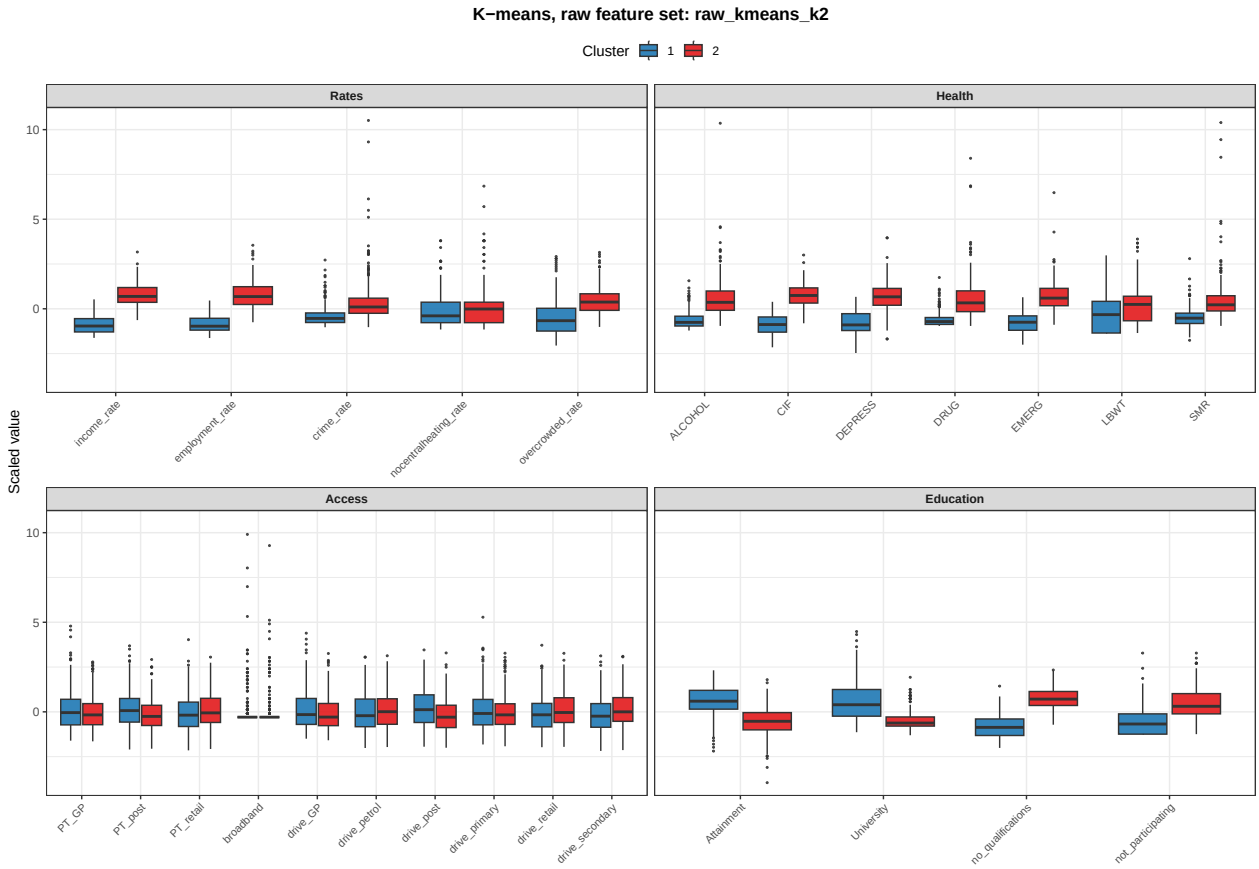



Table 7: Cluster sizes for raw kmeans k2

| cluster | n |
|---|---|
| 1 | 326 |
| 2 | 370 |




### 5.1.2 PC-based clustering

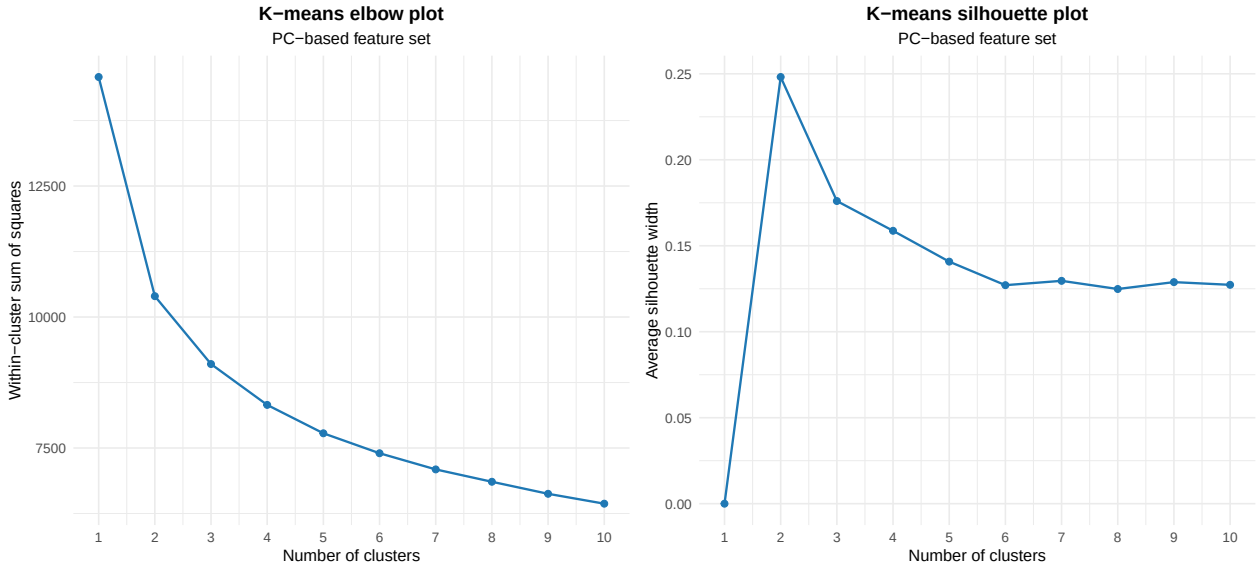
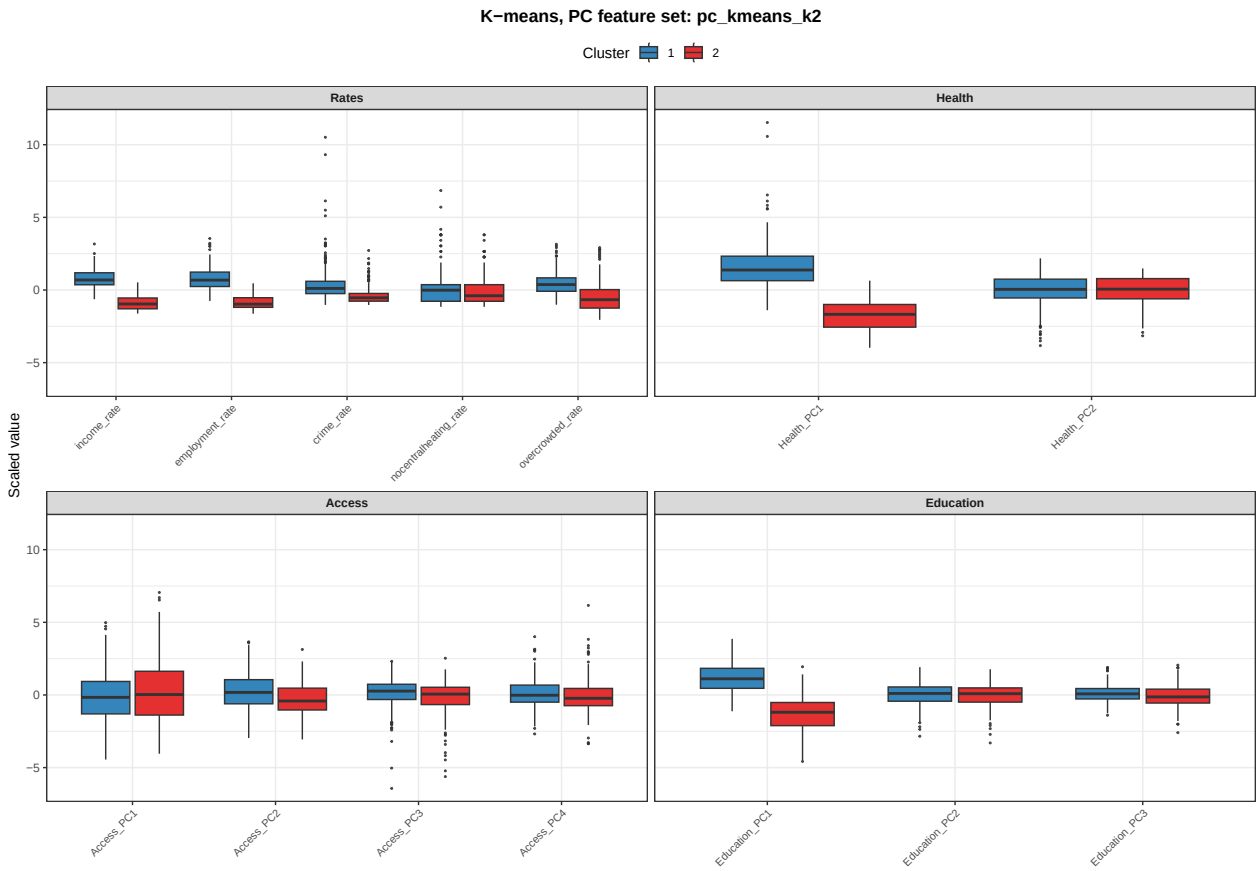



Table 8: Cluster sizes for pc kmeans k2

| cluster | n |
|---|---|
| 1 | 369 |
| 2 | 327 |



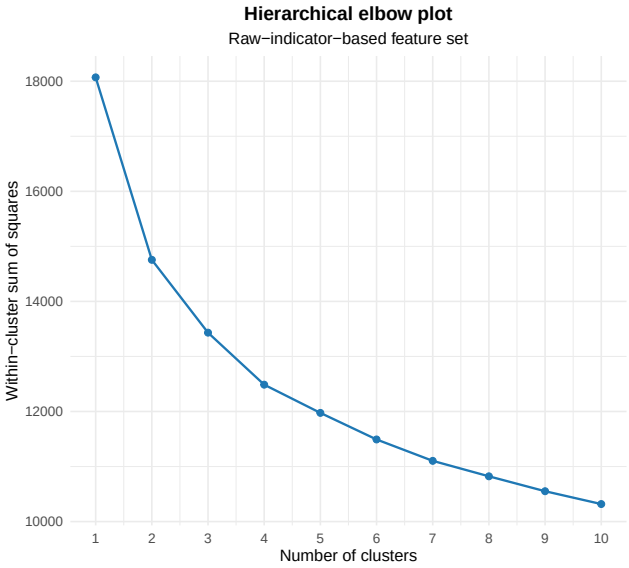

### 5.1.3 Comparison

Table 9: K-means comparison between raw and PC feature sets

| comparison | selected_k | best_silhouette_k | best_silhouette |
|---|---|---|---|
| raw | 2 | 2 | 0.205 |
| pc | 2 | 2 | 0.248 |

## 5.2 Hierarchical clustering

### 5.2.1 Raw-based clustering



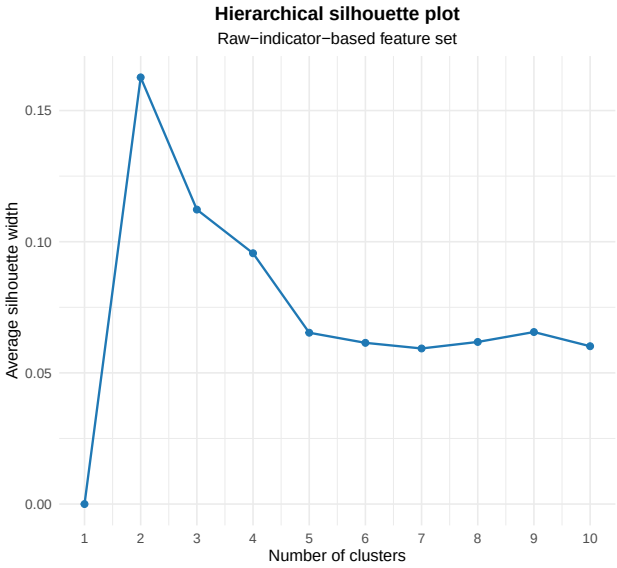
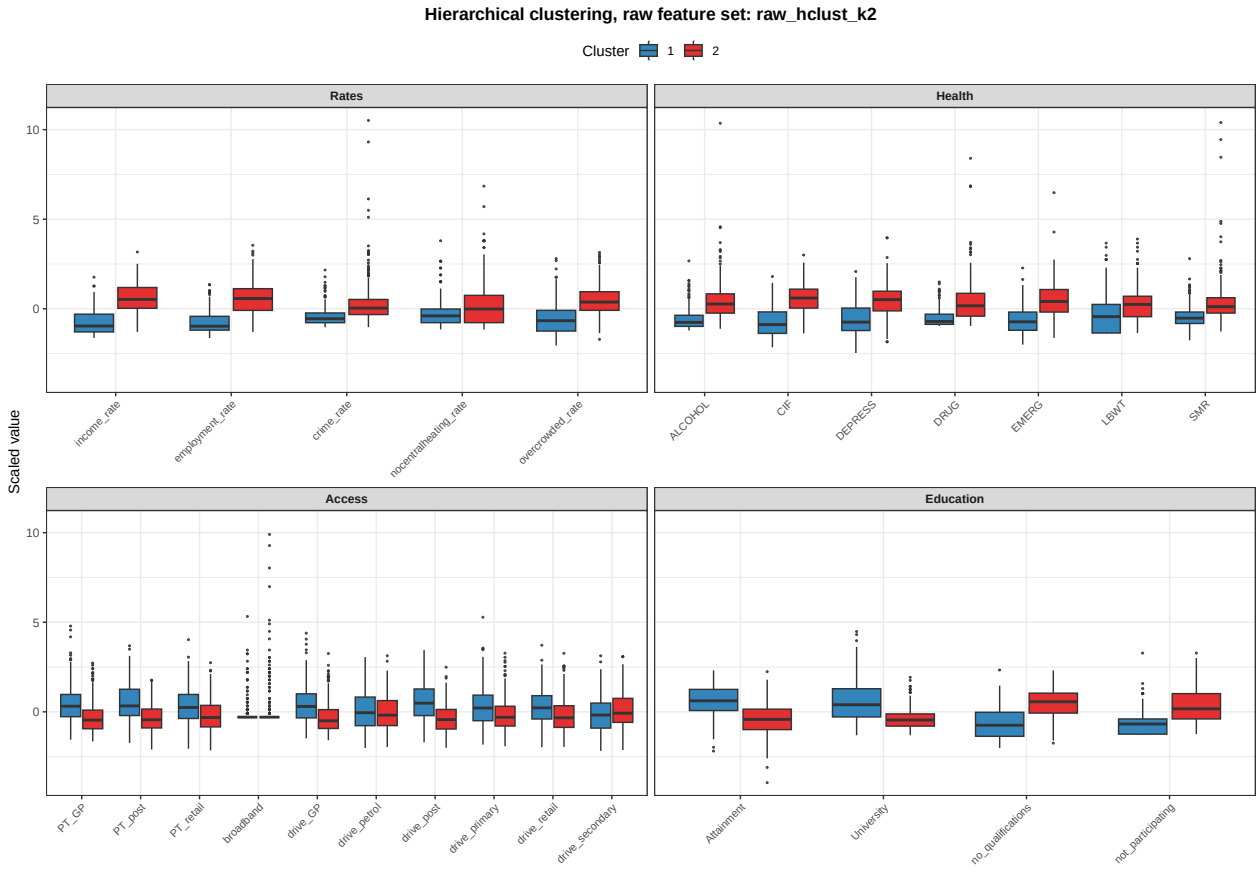




Table 10: Cluster sizes for raw hclust k2

| cluster | n |
|---|---|
| 1 | 288 |
| 2 | 408 |




### 5.2.2 PC-based clustering

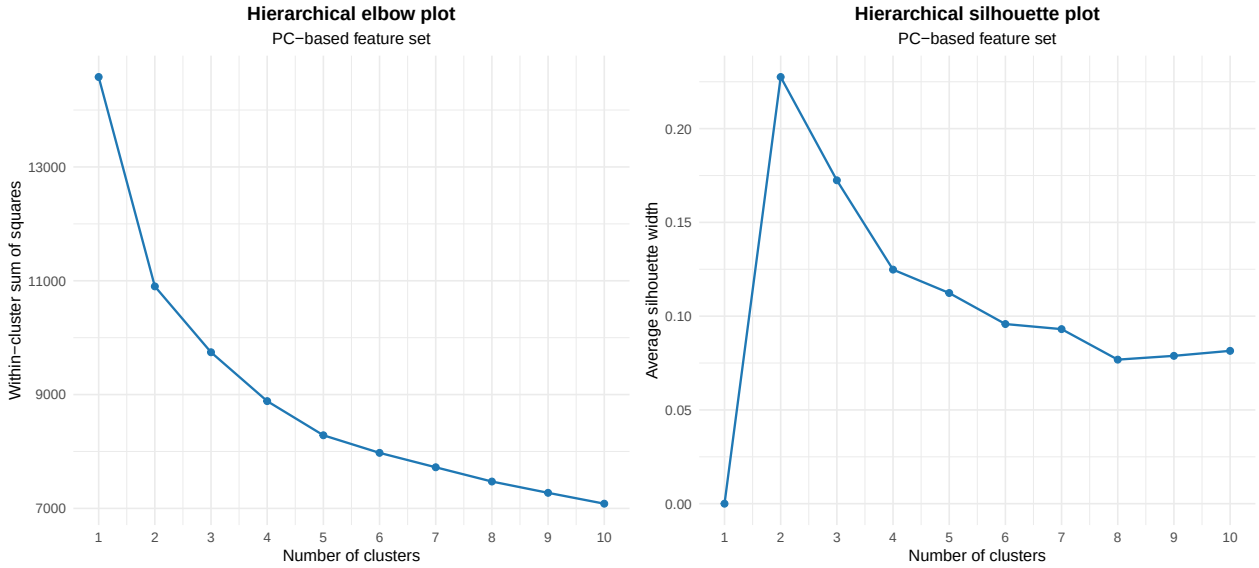
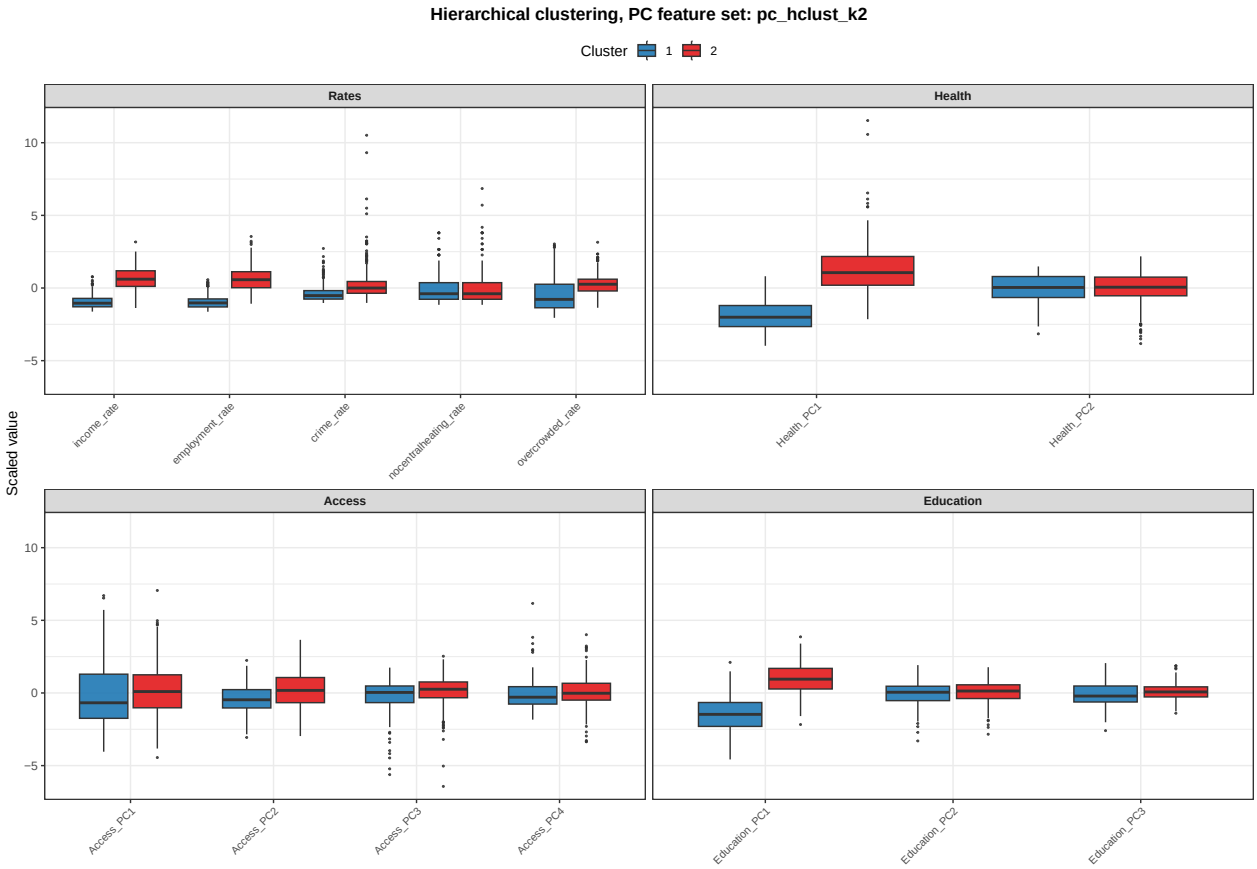



Table 11: Cluster sizes for pc hclust k2

| cluster | n |
|---|---|
| 1 | 274 |
| 2 | 422 |




### 5.2.3 Comparison

Table 12: Hierarchical clustering comparison between raw and PC feature sets

| comparison | selected_k | best_silhouette_k | best_silhouette |
|---|---|---|---|
| raw | 2 | 2 | 0.163 |
| pc | 2 | 2 | 0.228 |

## 5.3 C-means (sensitivity / uncertainty layer)

C-means is retained here as a **sensitivity / uncertainty layer** around the headline two-cluster K-means result. It is used to show which data zones are **core members** of each profile and which are **boundary cases**, not to reopen the choice of `k` and not to claim a sharp optimum in `m`.

Accordingly, `k` is held fixed at the chosen K-means value for each feature set, and the fuzziness exponent `m` is varied over a practical range (`params$cmeans_m_grid`, default `1.1` to `2.5`). The **reference/reporting value `m = 2`** is used for the headline C-means summaries, not as a strongly selected optimum. The main C-means diagnostics are therefore:

- mean maximum membership,
- proportion with maximum membership below 0.6 (main ambiguity threshold),
- proportion with maximum membership below 0.7 (stricter secondary threshold),
- ARI between adjacent `m` values,
- ARI between hardened C-means labels and the baseline K-means labels.

Silhouette and WSS are kept only as **secondary compactness summaries** of hardened assignments.

### 5.3.1 Raw-based clustering

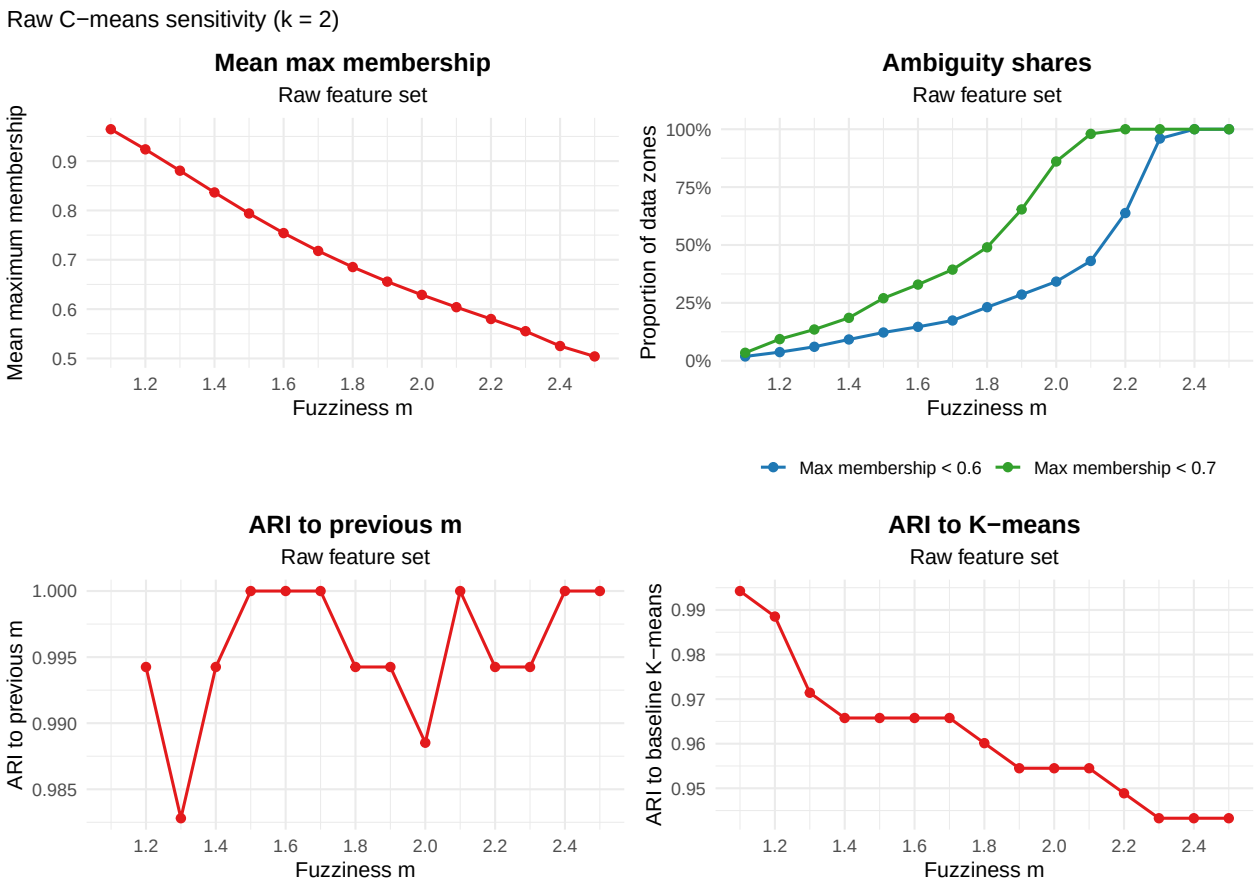
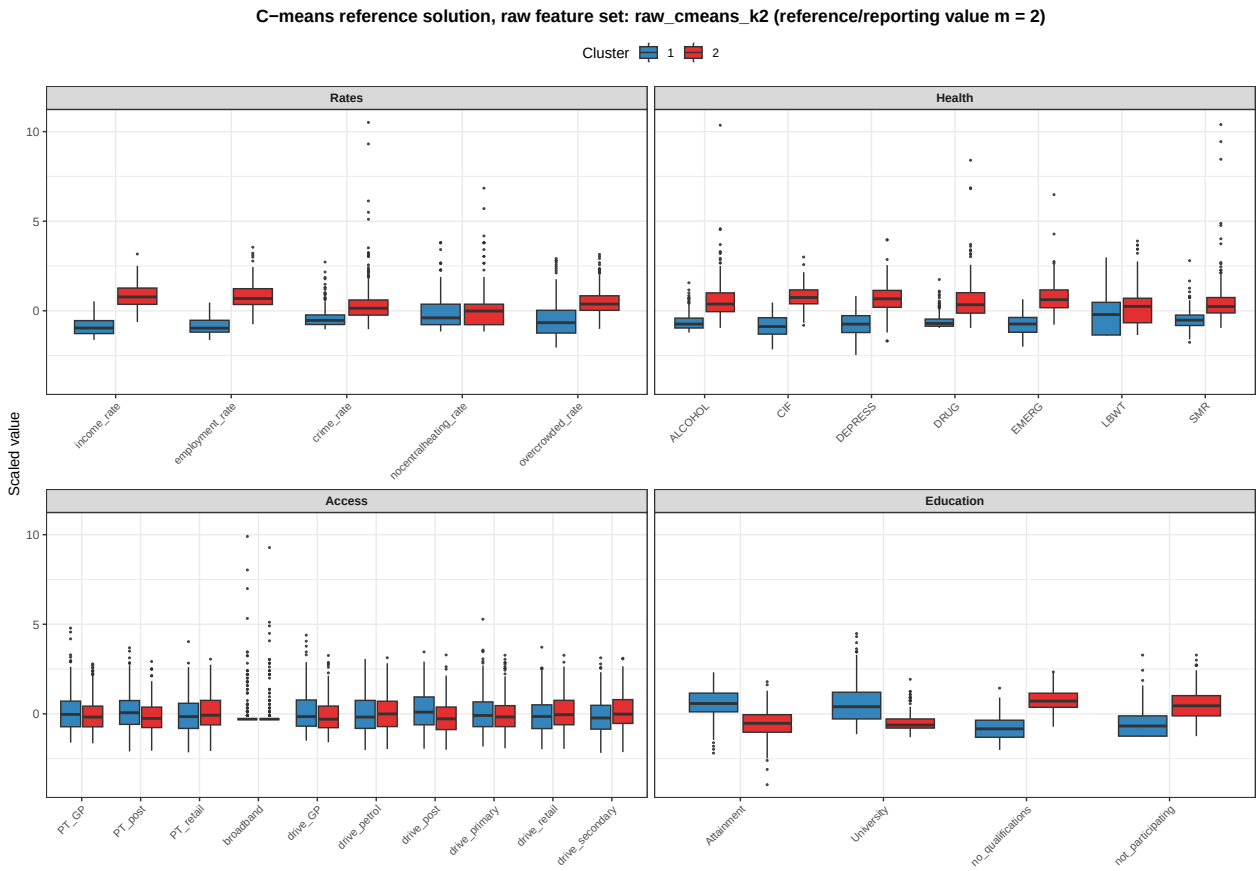
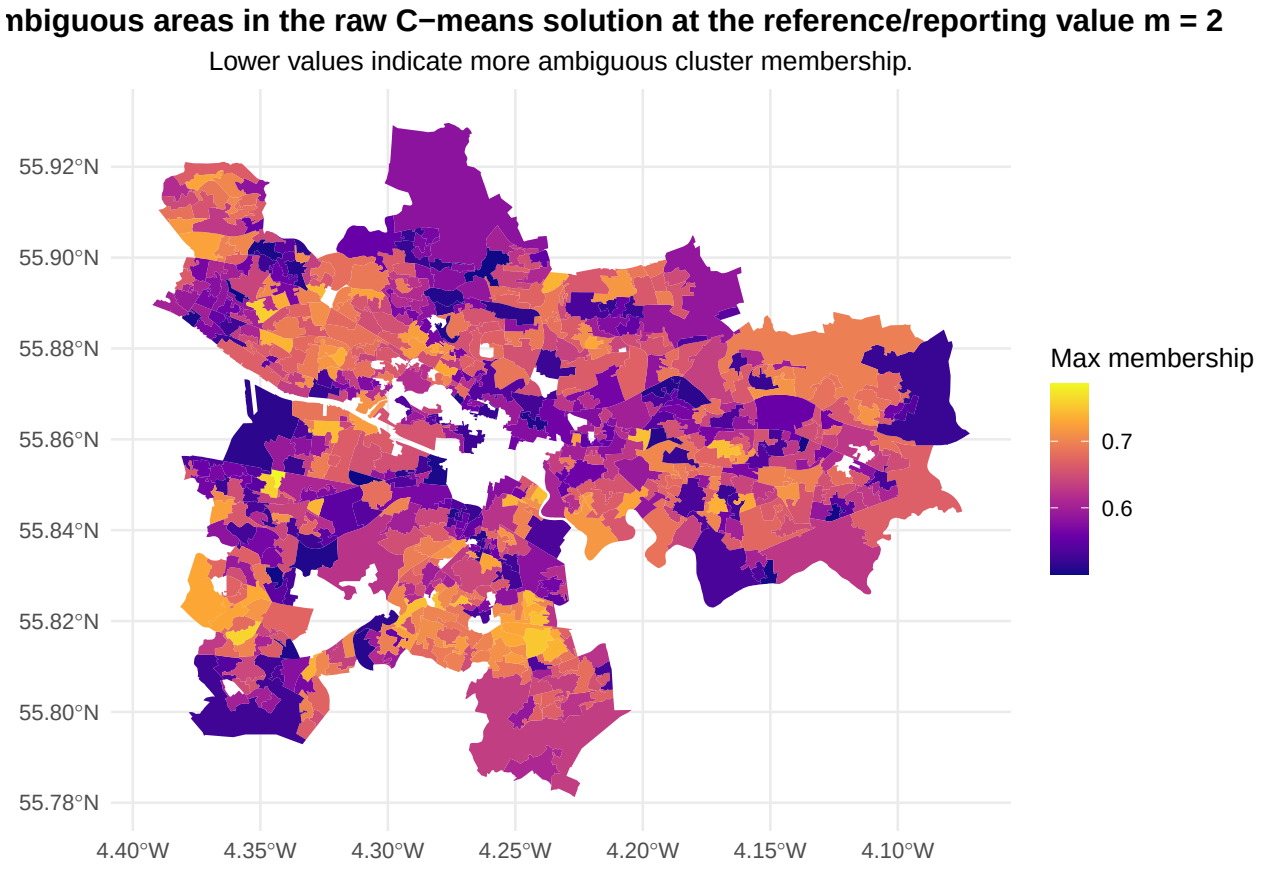



Table 13: Cluster sizes for raw cmeans k2

| cluster | n |
|---|---|
| 1 | 334 |
| 2 | 362 |


C−means reference solution, raw feature set: raw_cmeans_k2 (reference/reporting value m = 2)
Cluster 1 2
Rates
Health
Access
Education
Scaled value
10
5
0
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
ALCOHOL
CIF
DEPRESS
DRUG
EMERG
LBWT
SMR
PT_GP
PT_post
PT_retail
broadband
drive_GP
drive_petrol
drive_post
drive_primary
drive_retail
drive_secondary
Attainment
University
no_qualifications
not_participating


nbiguous areas in the raw C−means solution at the reference/reporting value m = 2
Lower values indicate more ambiguous cluster membership.
55.92°N
55.90°N
55.88°N
55.86°N
55.84°N
55.82°N
55.80°N
55.78°N
4.40°W
4.35°W
4.30°W
4.25°W
4.20°W
4.15°W
4.10°W
Max membership
0.7
0.6

### 5.3.2 PC-based clustering

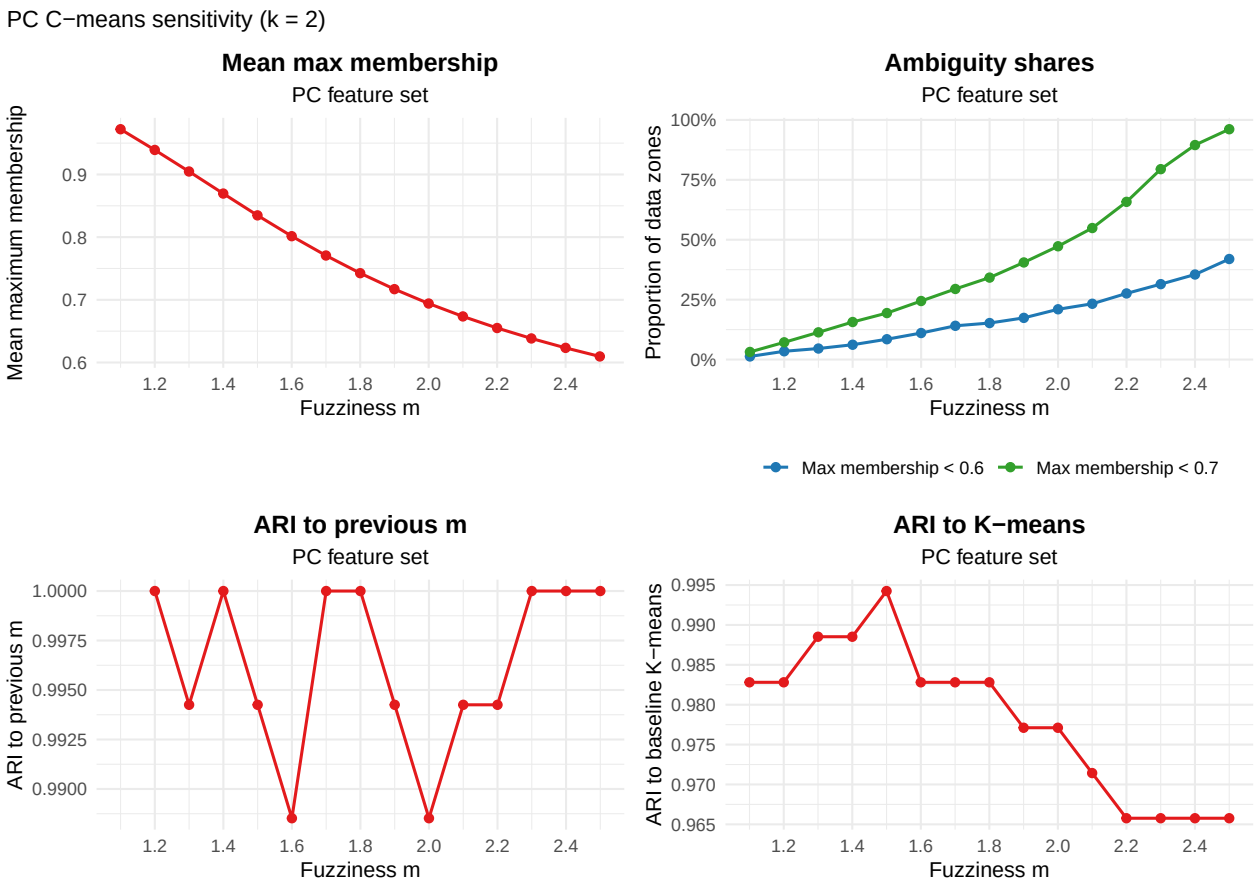
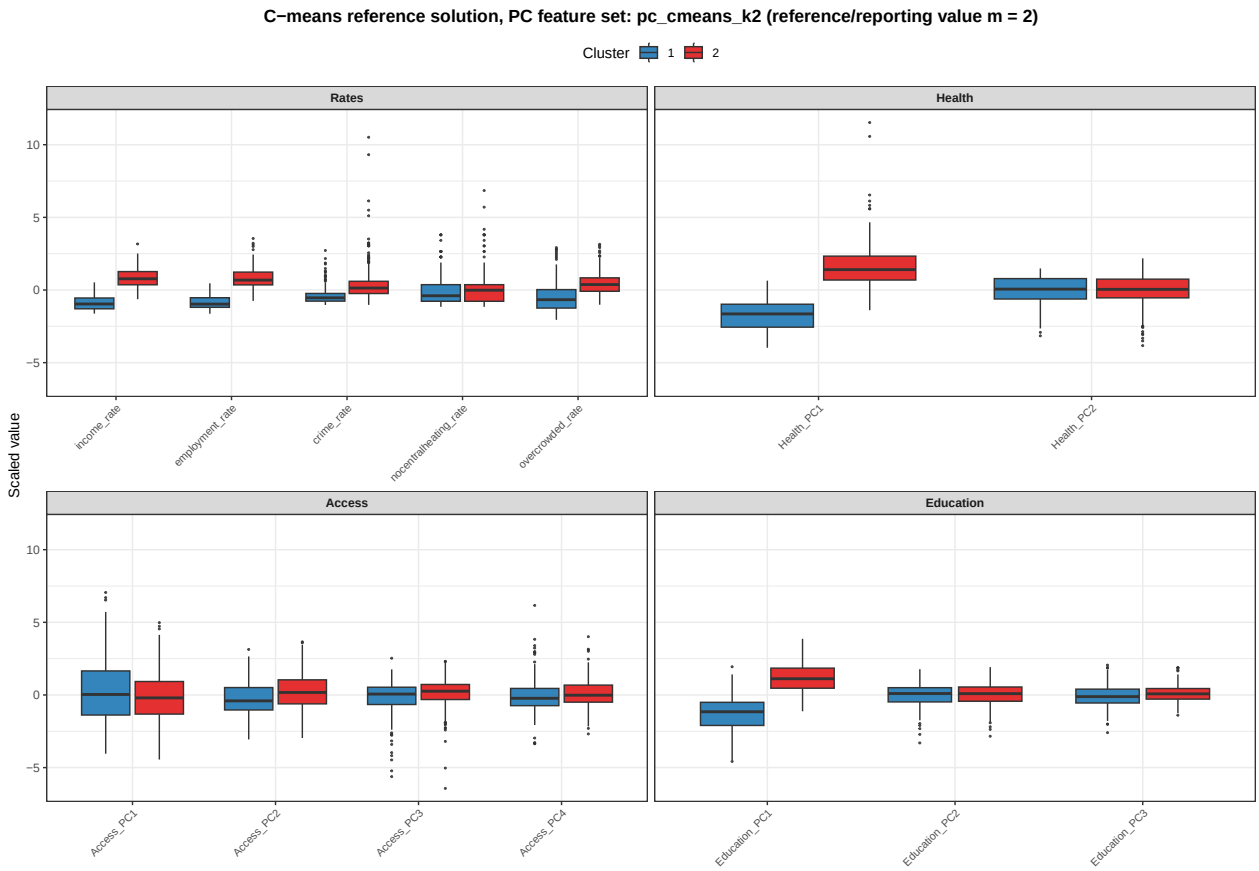
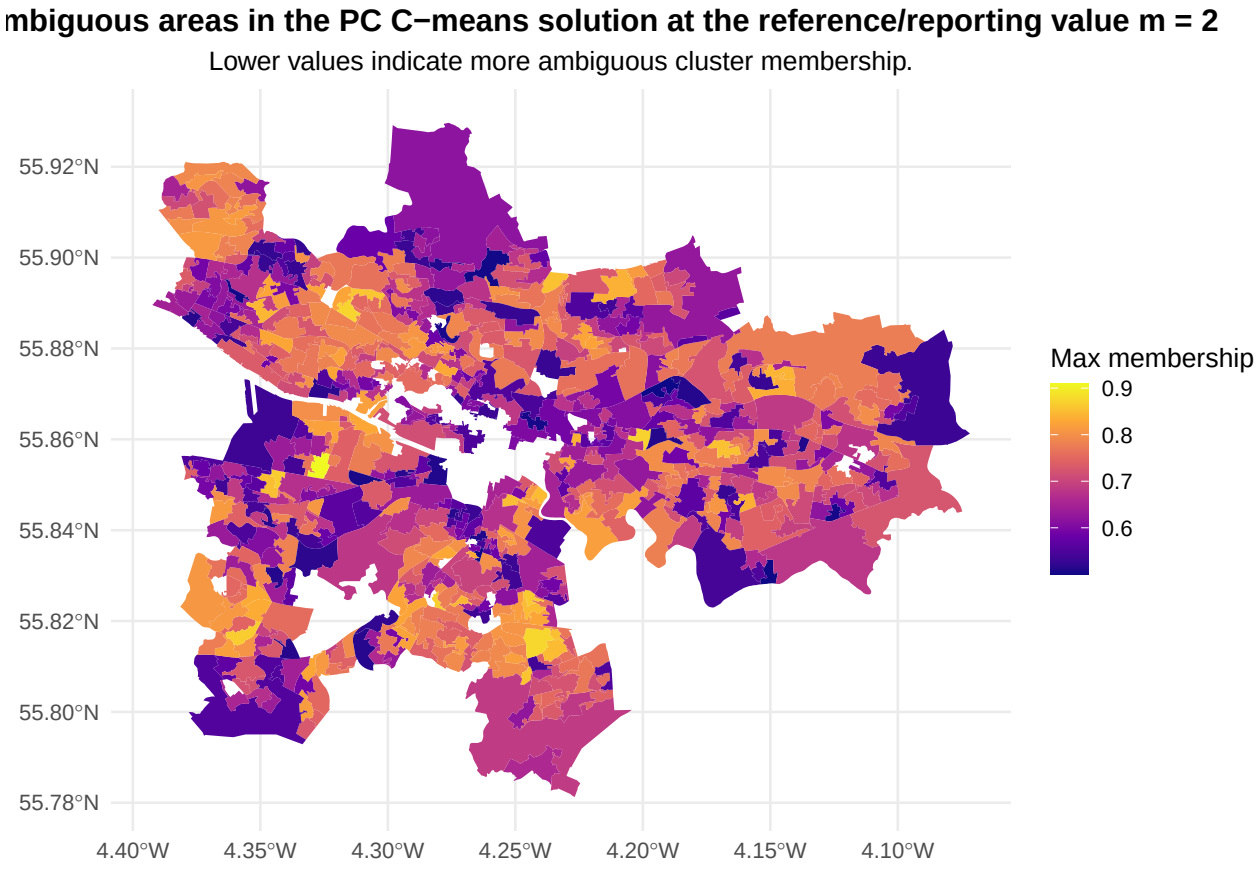



Table 14: Cluster sizes for pc cmeans k2

| cluster | n |
|---|---|
| 1 | 331 |
| 2 | 365 |


C−means reference solution, PC feature set: pc_cmeans_k2 (reference/reporting value m = 2)
Cluster
1
2
Rates
Health
Access
Education
Scaled value
10
5
0
−5
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
Health_PC1
Health_PC2
Access_PC1
Access_PC2
Access_PC3
Access_PC4
Education_PC1
Education_PC2
Education_PC3



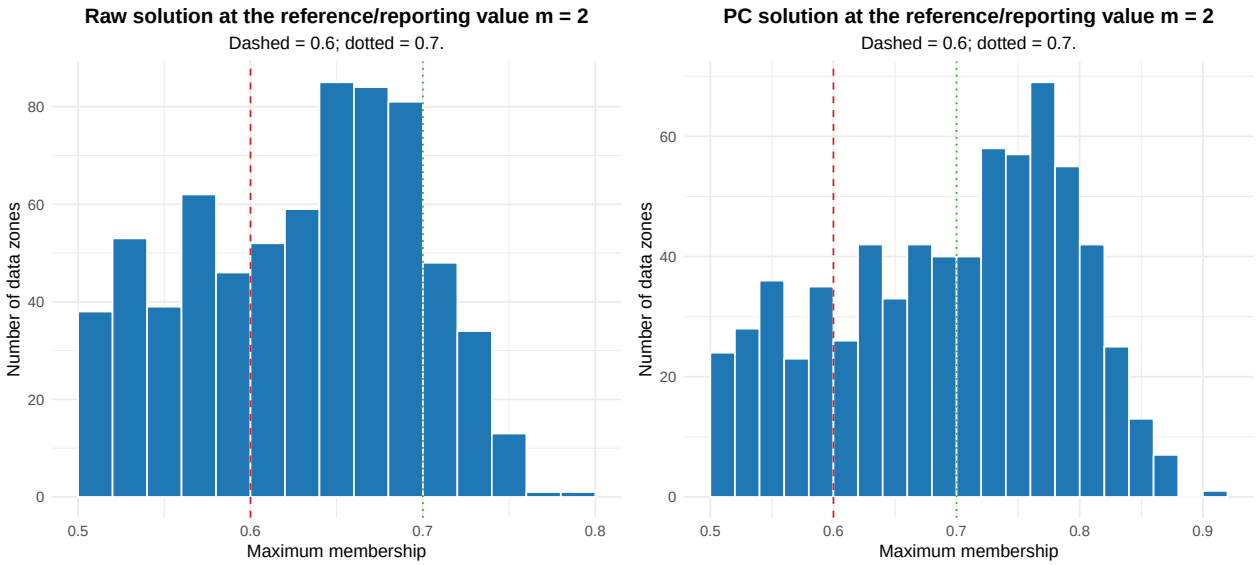

### 5.3.3 Reference-solution comparison

For this comparison, `k` is fixed at 2 and the **reference/reporting value `m = 2`** is used for both feature sets.

Table 15: Reference C-means summary by feature set

| Feature set | k | Reference/reporting value m | Summary | Value |
|---|---|---|---|---|
| PC | 2 | 2 | Mean max membership | 0.694 |
| PC | 2 | 2 | Prop. max < 0.6 | 0.210 |
| PC | 2 | 2 | Prop. max < 0.7 | 0.473 |
| PC | 2 | 2 | ARI to previous m | 0.989 |
| PC | 2 | 2 | ARI to K-means | 0.977 |
| PC | 2 | 2 | Silhouette (hard labels) | 0.248 |
| Raw | 2 | 2 | Mean max membership | 0.629 |
| Raw | 2 | 2 | Prop. max < 0.6 | 0.342 |
| Raw | 2 | 2 | Prop. max < 0.7 | 0.861 |
| Raw | 2 | 2 | ARI to previous m | 0.989 |
| Raw | 2 | 2 | ARI to K-means | 0.954 |
| Raw | 2 | 2 | Silhouette (hard labels) | 0.205 |

At the reference/reporting value `m = 2`, the PC-based two-cluster solution is sharper, while the raw solution contains more boundary cases.




## 5.4 Finite mixture models (secondary model-based comparator)

Finite mixture models are kept here as a **secondary model-based comparator**, not as co-equal evidence for the same two-cluster story. In these data, the BIC-preferred component count can differ materially from `k = 2`, so the two-component FMM should be read as a **constrained comparator for consistency**, not as independent support for the headline two-cluster conclusion.

BIC and ICL are therefore the main FMM diagnostics. Silhouette is shown only as a **secondary compactness summary** of the hardened assignments.

### 5.4.1 Raw-based clustering

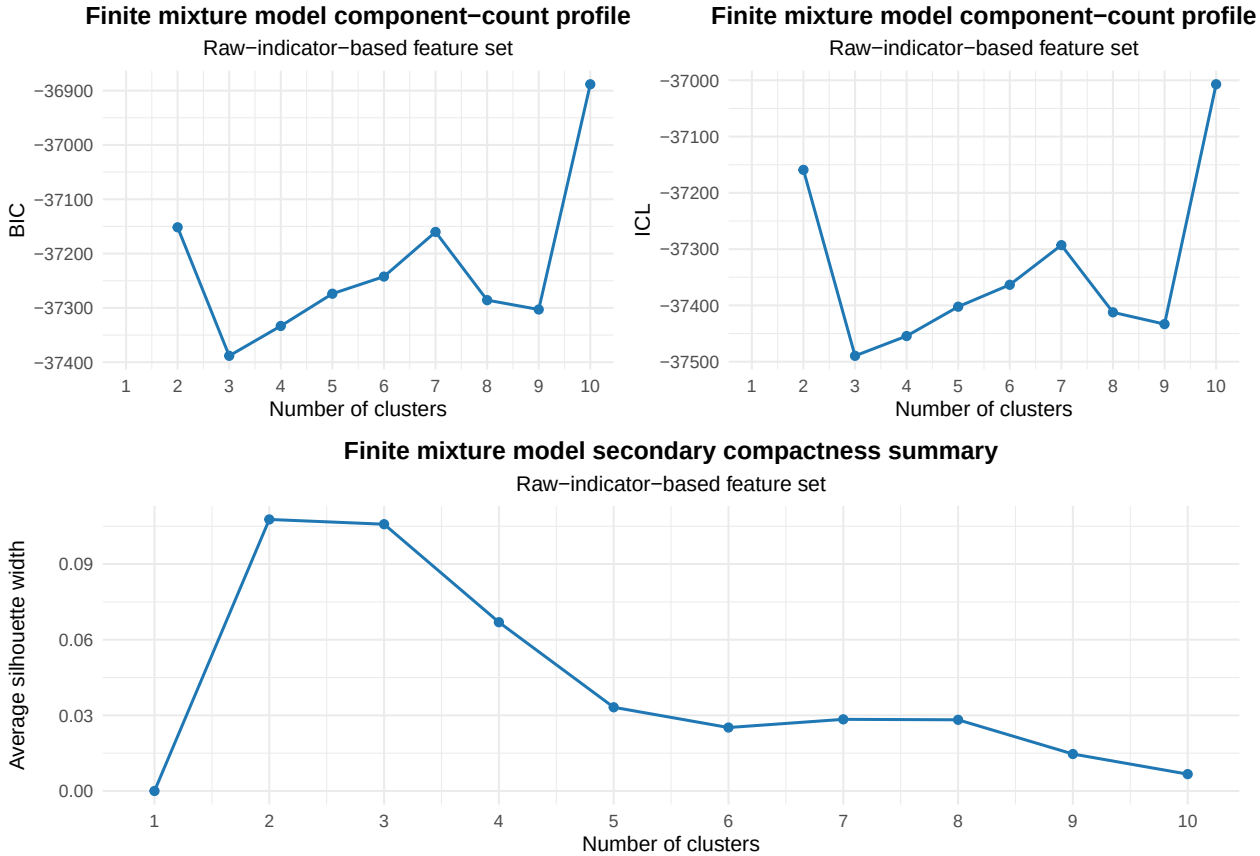
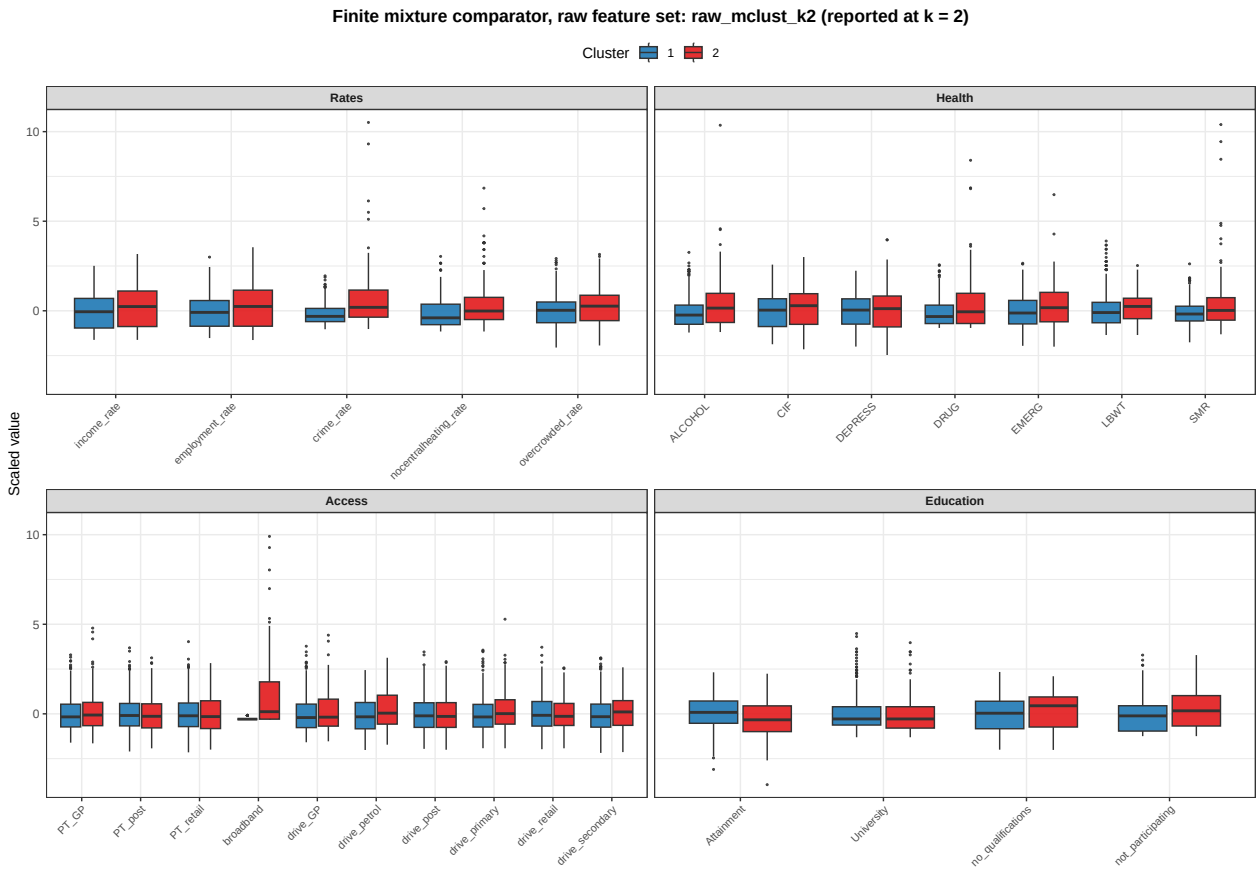



Table 16: Cluster sizes for constrained comparator raw mclust k2

| cluster | n |
|---|---|
| 1 | 532 |
| 2 | 164 |


Finite mixture comparator, raw feature set: raw_mclust_k2 (reported at k = 2)
Cluster 1 2
Rates
Health
Access
Education
Scaled value
10
5
0
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
ALCOHOL
CIF
DEPRESS
DRUG
EMERG
LBWT
SMR
PT_GP
PT_post
PT_retail
broadband
drive_GP
drive_petrol
drive_post
drive_primary
drive_retail
drive_secondary
Attainment
University
no_qualifications
not_participating

### 5.4.2 PC-based clustering

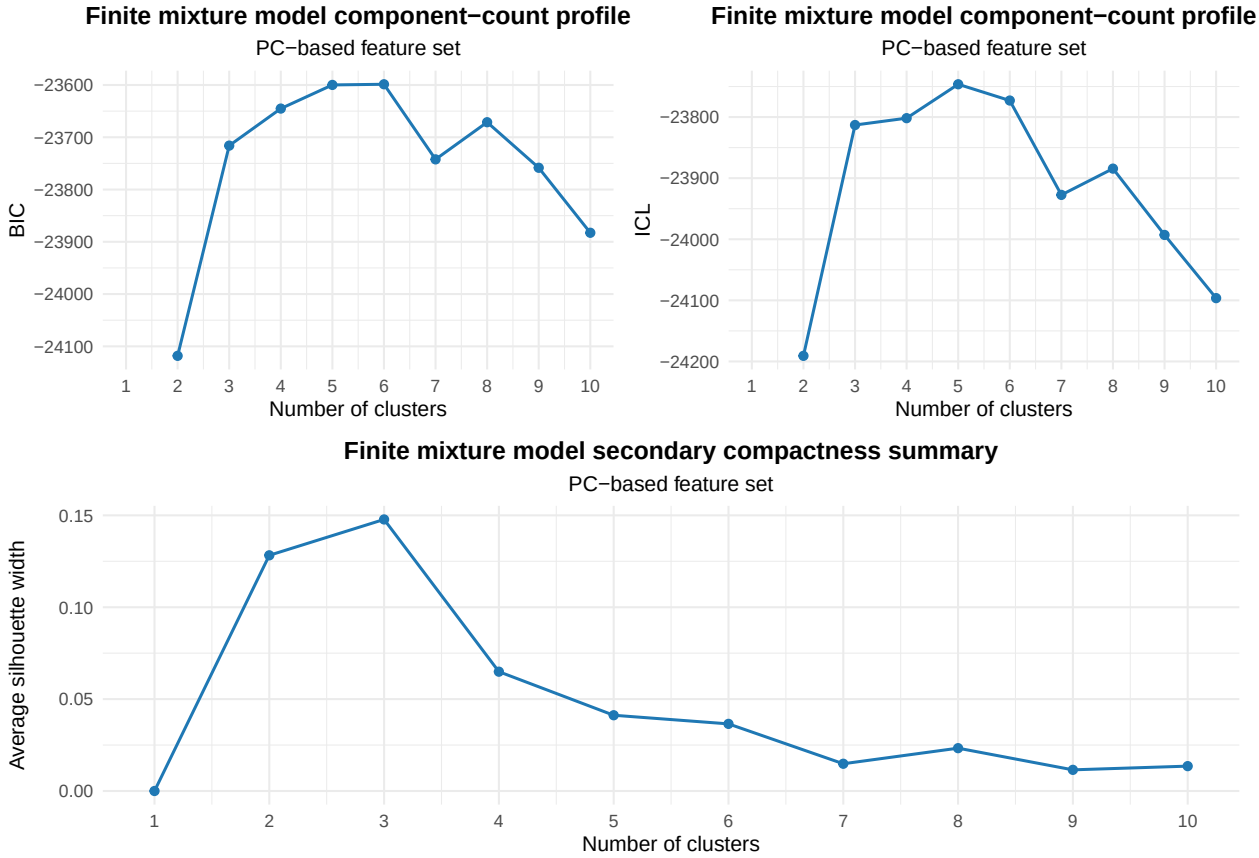
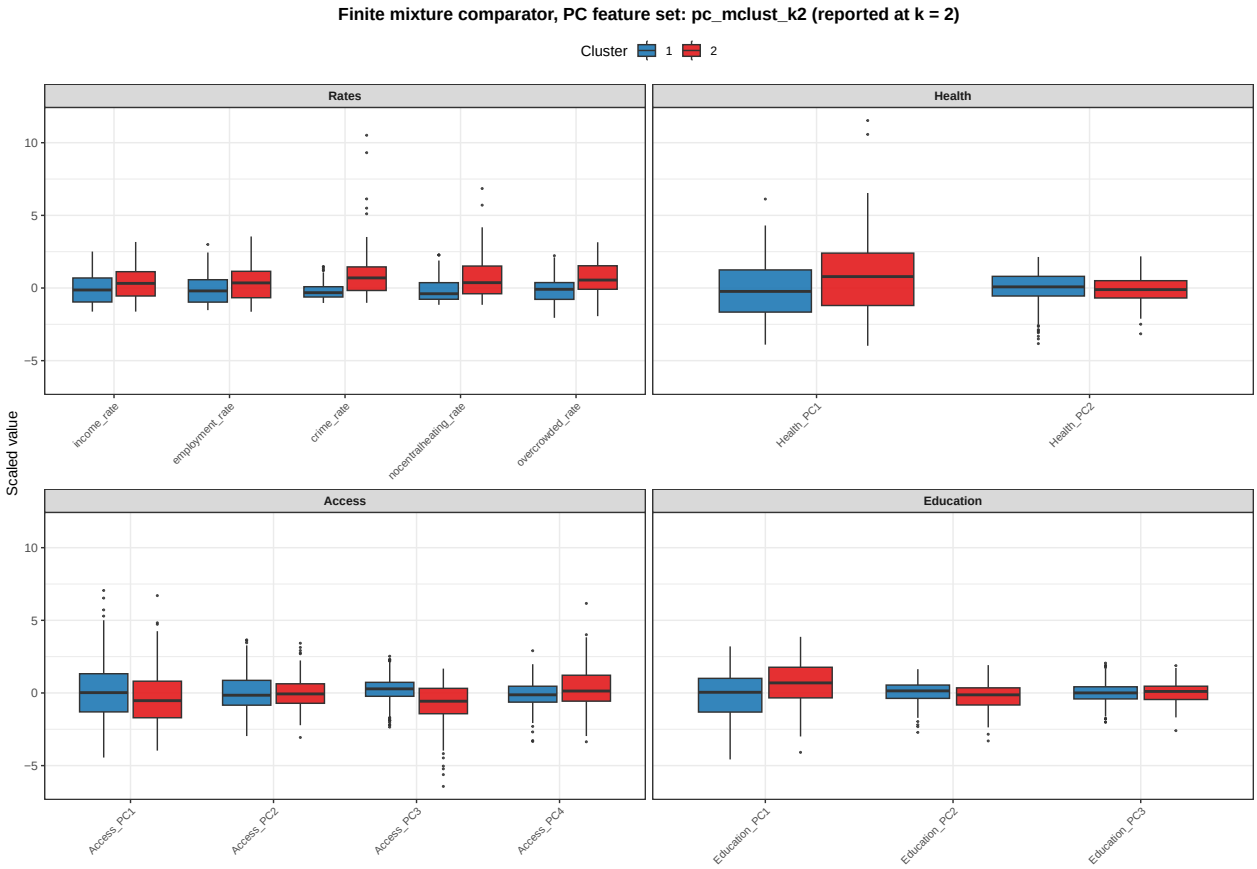



Table 17: Cluster sizes for constrained comparator pc mclust k2

| cluster | n |
|---|---|
| 1 | 548 |
| 2 | 148 |




### 5.4.3 Comparator summary

Table 18: Finite mixture comparator summary between raw and PC feature sets

| comparison | reporting_k | best_bic_k | best_bic | best_silhouette_k | best_silhouette |
|---|---|---|---|---|---|
| raw | 2 | 10 | -36888.32 | 2 | 0.108 |
| pc | 2 | 6 | -23598.64 | 3 | 0.148 |

# 6 Comparison of selected clustering results

This final comparison keeps the roles of the methods distinct: **K-means remains the headline partitioning result**, **C-means is an uncertainty overlay**, and **FMM is a secondary model-based comparator**.

## 6.1 Diagnostic summaries

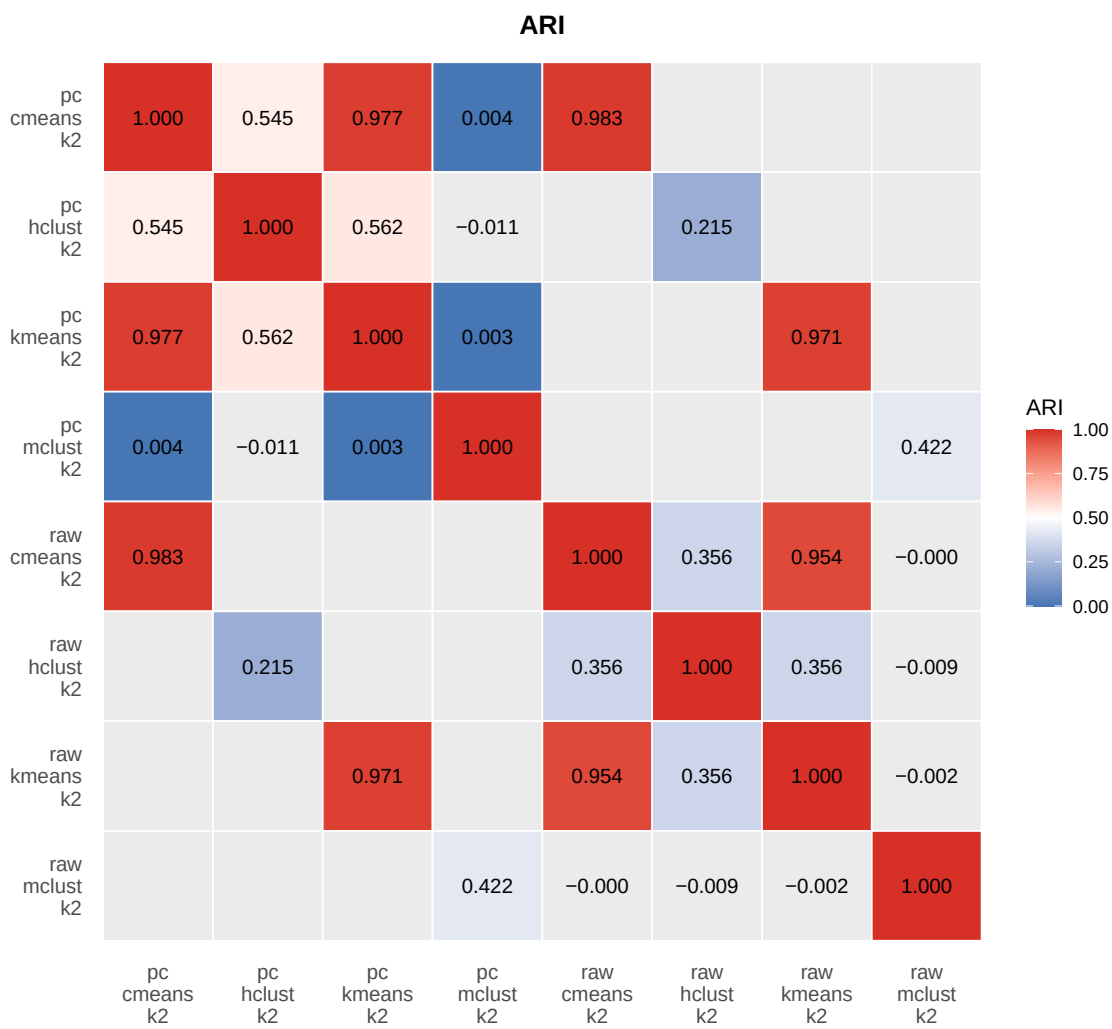
## 6.2 Similarity cross-tables

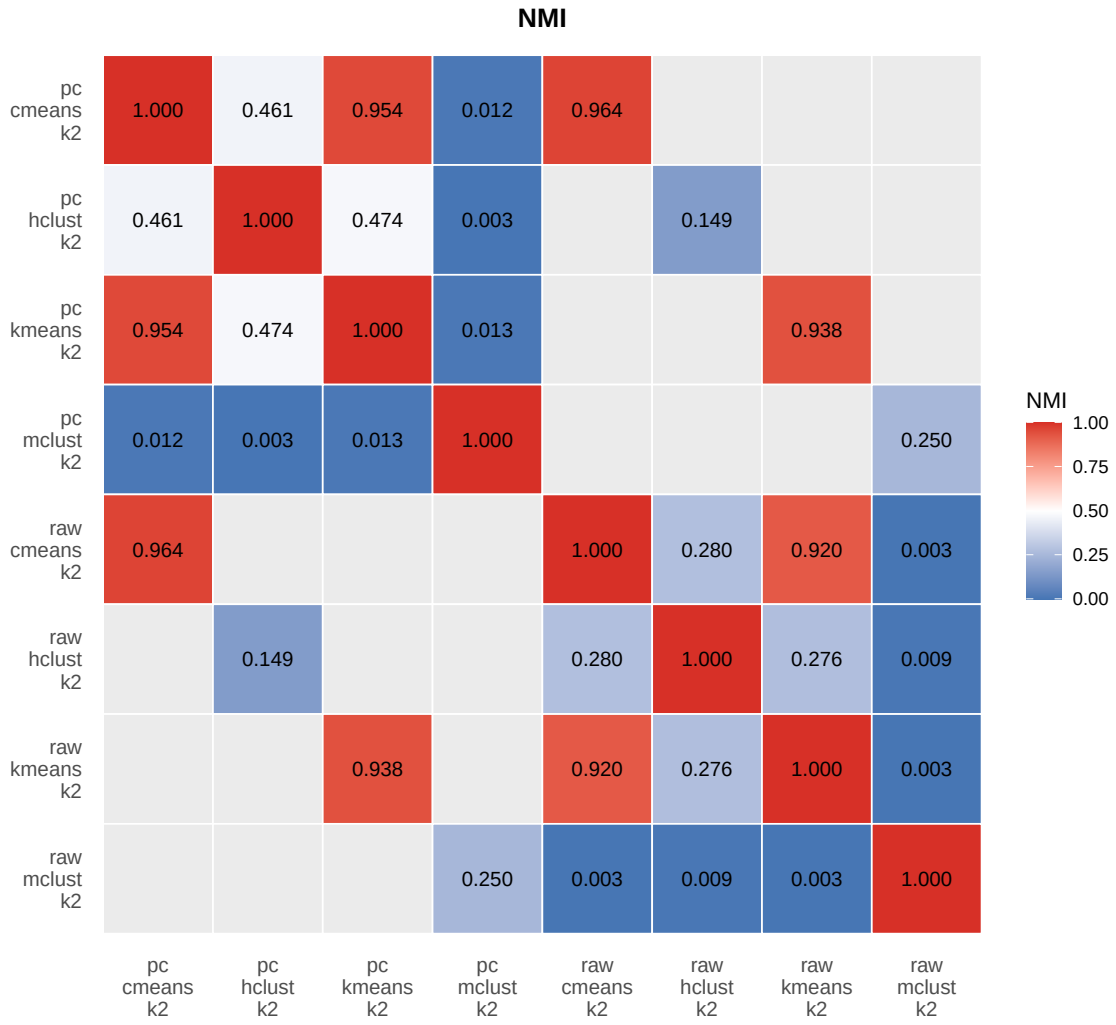
Table 19: Selected pairwise similarity metrics

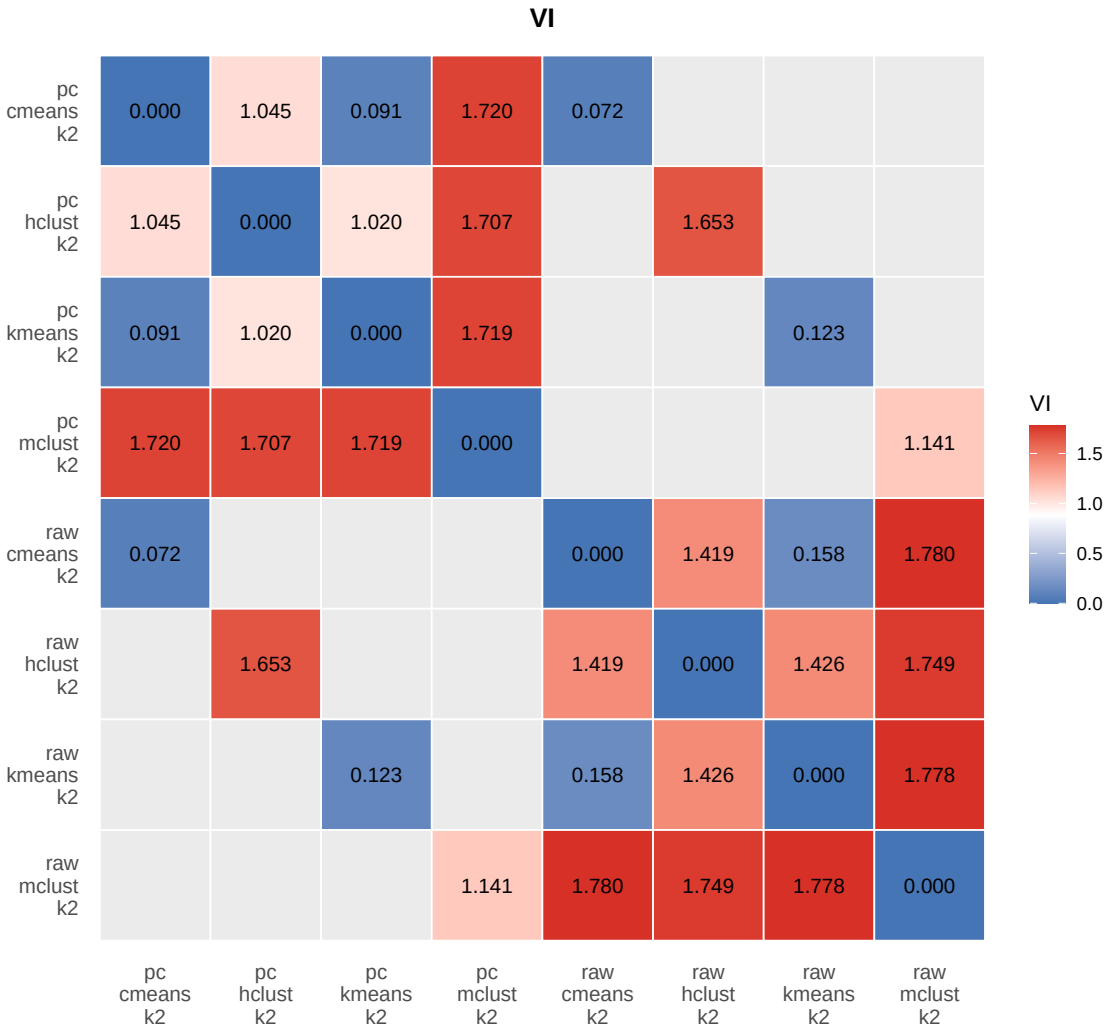
| clustering_a | clustering_b | ARI | NMI | VI |
|---|---|---|---|---|
| raw_cmeans_k2 | pc_cmeans_k2 | 0.983 | 0.964 | 0.072 |
| pc_cmeans_k2 | pc_kmeans_k2 | 0.977 | 0.954 | 0.091 |

| | | | | |
|---|---|---|---|---|
| raw_kmeans_k2 | pc_kmeans_k2 | 0.971 | 0.938 | 0.123 |
| raw_cmeans_k2 | raw_kmeans_k2 | 0.954 | 0.920 | 0.158 |
| pc_hclust_k2 | pc_kmeans_k2 | 0.562 | 0.474 | 1.020 |
| pc_cmeans_k2 | pc_hclust_k2 | 0.545 | 0.461 | 1.045 |
| raw_mclust_k2 | pc_mclust_k2 | 0.422 | 0.250 | 1.141 |
| raw_cmeans_k2 | raw_hclust_k2 | 0.356 | 0.280 | 1.419 |
| raw_hclust_k2 | raw_kmeans_k2 | 0.356 | 0.276 | 1.426 |
| raw_hclust_k2 | pc_hclust_k2 | 0.215 | 0.149 | 1.653 |
| pc_cmeans_k2 | pc_mclust_k2 | 0.004 | 0.012 | 1.720 |
| pc_kmeans_k2 | pc_mclust_k2 | 0.003 | 0.013 | 1.719 |
| raw_cmeans_k2 | raw_mclust_k2 | 0.000 | 0.003 | 1.780 |
| raw_kmeans_k2 | raw_mclust_k2 | -0.002 | 0.003 | 1.778 |
| raw_hclust_k2 | raw_mclust_k2 | -0.009 | 0.009 | 1.749 |
| pc_hclust_k2 | pc_mclust_k2 | -0.011 | 0.003 | 1.707 |

The square diagrams below are shown one metric at a time so that the labels and values remain readable in the knitted report.

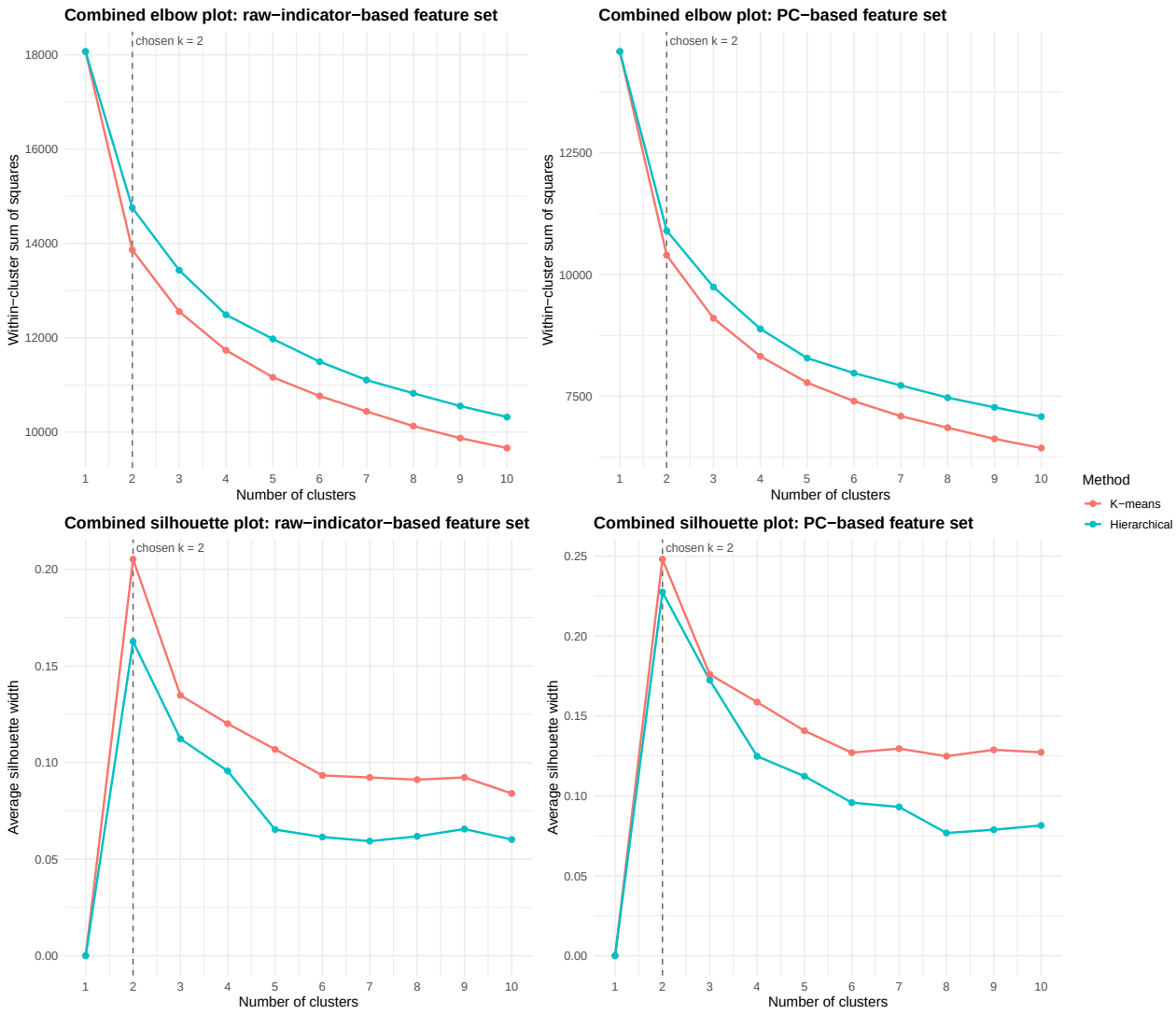



NMI
pc cmeans k2
pc hclust k2
pc kmeans k2
pc mclust k2
raw cmeans k2
raw hclust k2
raw kmeans k2
raw mclust k2
1.000 0.461 0.954 0.012 0.964
0.461 1.000 0.474 0.003 0.149
0.954 0.474 1.000 0.013 0.938
0.012 0.003 0.013 1.000 0.250
0.964 1.000 0.280 0.920 0.003
0.149 0.280 1.000 0.276 0.009
0.938 0.920 0.276 1.000 0.003
0.250 0.003 0.009 0.003 1.000
NMI
1.00
0.75
0.50
0.25
0.00




## 6.3 Primary combined diagnostics for the headline hard-partitioning methods

The combined comparison below is intentionally limited to **K-means** and **hierarchical clustering**, which carry the main hard-partition story. C-means and FMM are summarised in their own sections above; for those methods, silhouette and WSS are secondary diagnostics rather than the basis for the main conclusion.

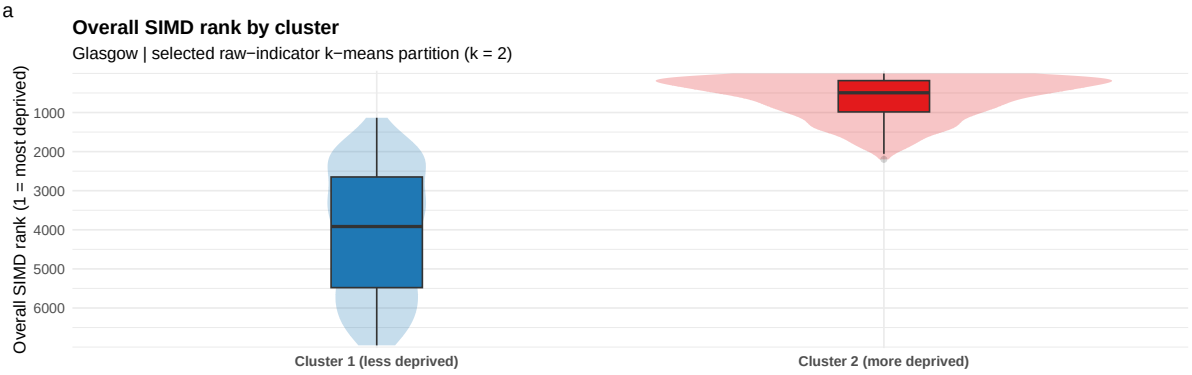



## 6.4 External anchoring to overall SIMD rank

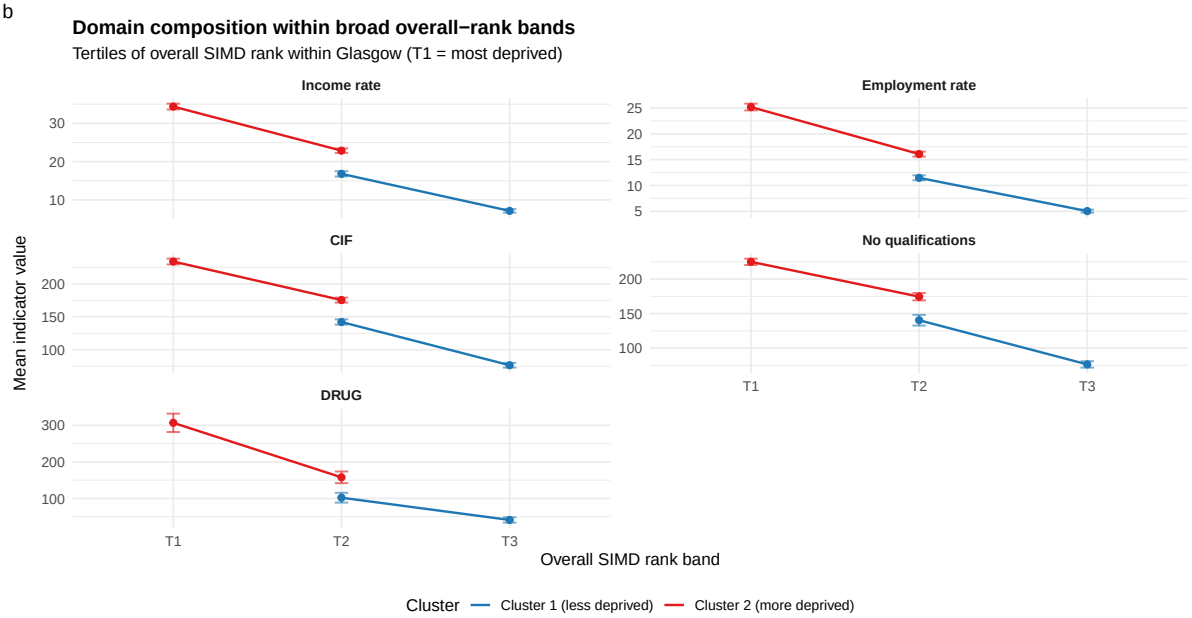
The two-panel figure below anchors the selected cluster solution back to overall SIMD position. Panel (a) shows whether the selected clusters track overall deprivation rank. Panel (b) uses a tightened five-indicator set and shows 95% confidence intervals around within-tertile cluster means, so that differences within broad overall-rank bands read more clearly as evidence rather than as an exploratory diagnostic.

Relationship between selected cluster membership and overall SIMD rank in Glasgow

a

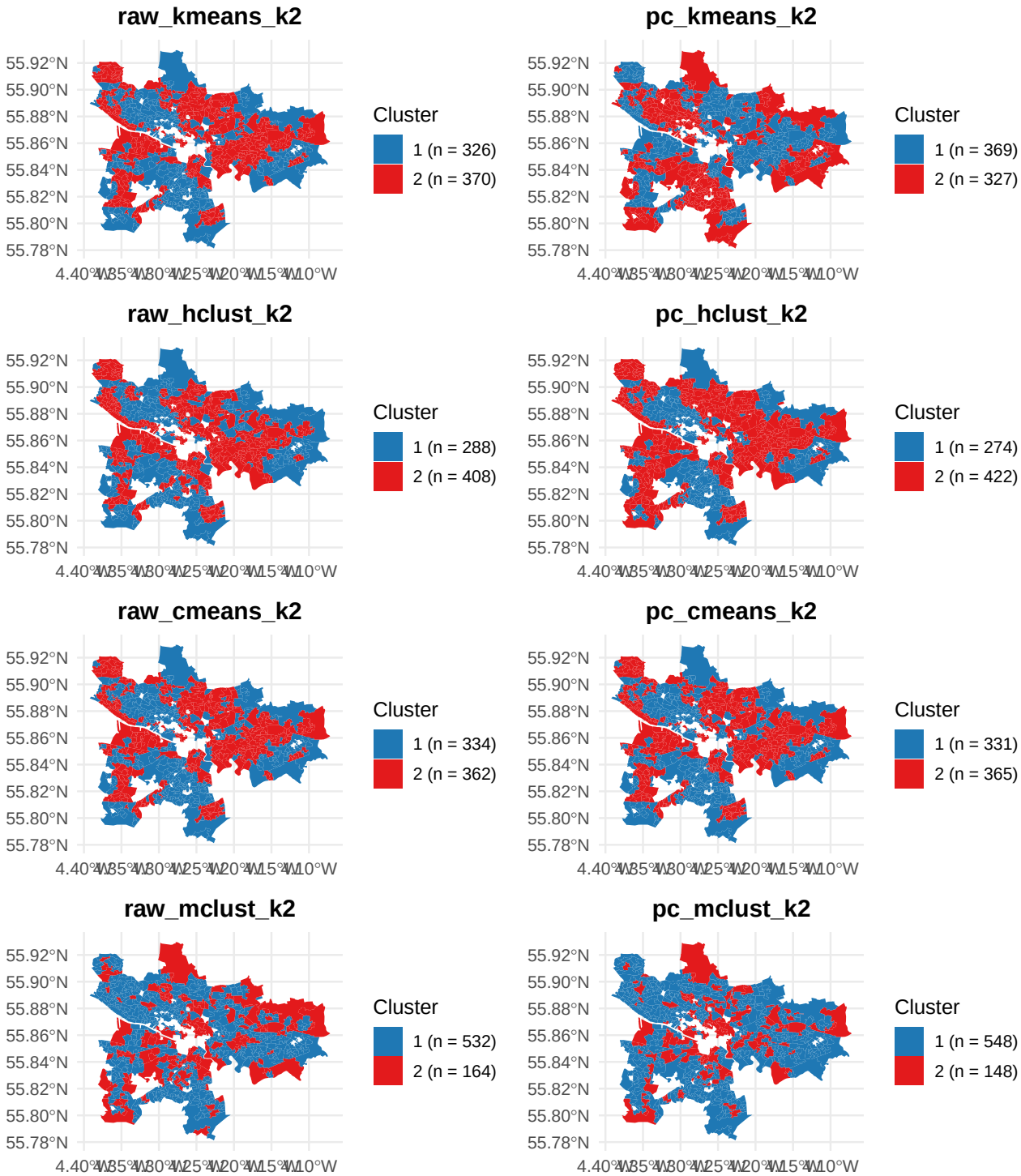



b



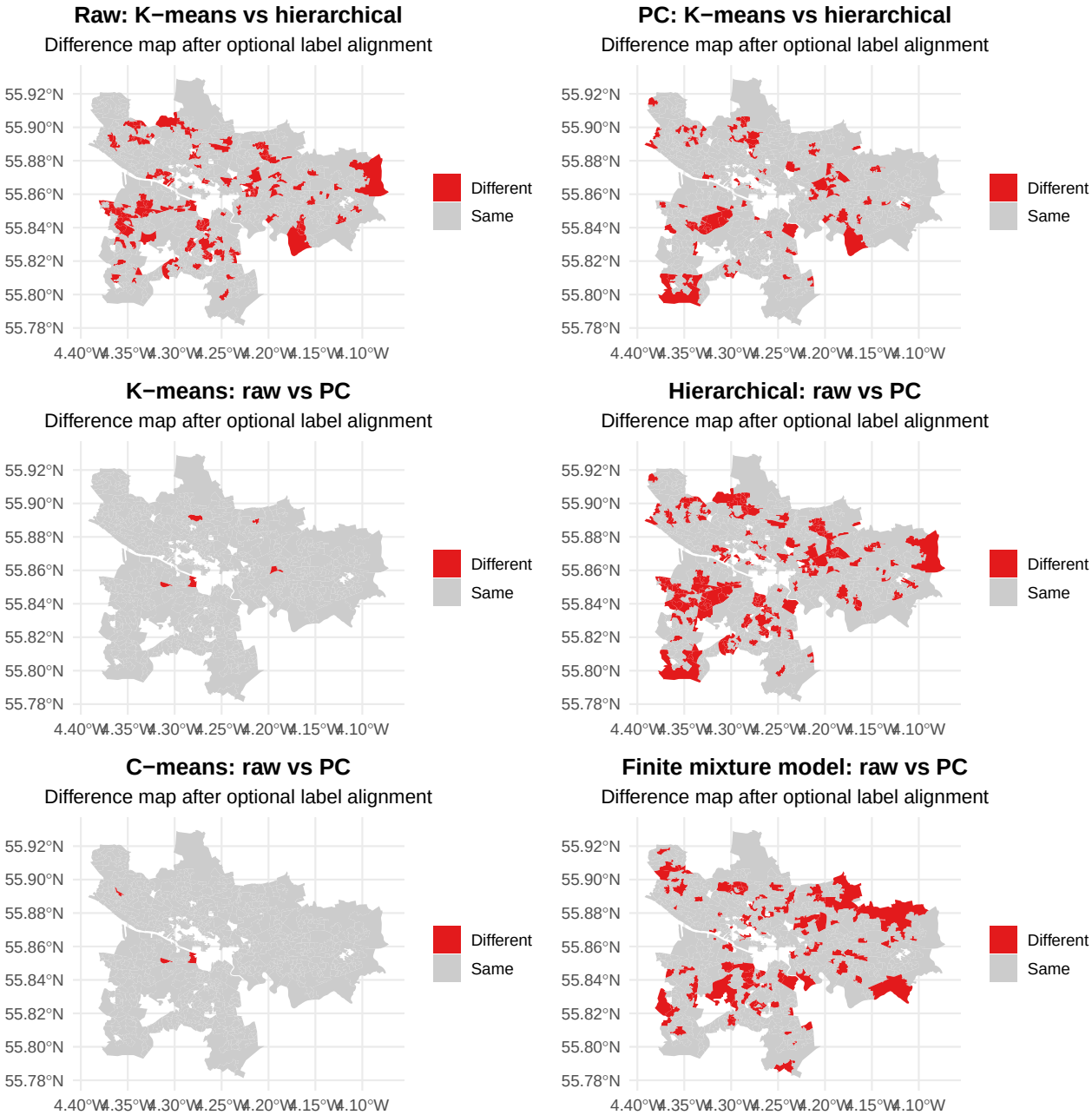
## 6.5 Maps for selected clustering results

Maps for selected clustering results



## 6.6 Difference maps

Difference maps for selected clustering comparisons




# 7 Outputs passed to the vine document

The next document should read:

- glasgow_analysis_base.csv
- glasgow_feature_set_raw.csv
- glasgow_feature_set_pc.csv
- glasgow_cluster_assignments.csv

If a reference C-means solution is carried forward as an uncertainty-aware comparator, it should also read:

- glasgow_cmeans_memberships_raw.csv
- glasgow_cmeans_memberships_pc.csv

These C-means membership files contain:

- `DataZone`
- `cluster_1_membership`
- `cluster_2_membership`
- `max_membership`
- `hard_label`
- `core_borderline_flag`

The chosen clustering result for vine fitting is then selected by its column name, for example:

- `raw_kmeans_k2`
- `pc_hclust_k2`
- `raw_cmeans_k2`
- `pc_mclust_k2`

A useful next-stage sensitivity check is to fit vines on **all members** first and then refit on a stricter **core-only subset** defined from `max_membership` (for example >= 0.8) for the reference C-means solution. The updated vine document supports this directly through optional C-means membership inputs and a core-membership threshold parameter.

# 8 Session info

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS 15.7.4
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] reshape2_1.4.4   kableExtra_1.4.0 knitr_1.49       salso_0.3.53
##  [5] aricode_1.0.3    mclust_6.1.1     e1071_1.7-16     patchwork_1.3.2
##  [9] factoextra_1.0.7 cluster_2.1.6    sf_1.0-21        lubridate_1.9.4
## [13] forcats_1.0.1    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
## [17] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_4.0.0
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       xfun_0.53          ggrepel_0.9.6      lattice_0.22-6
##  [5] tzdb_0.4.0         vctrs_0.6.5        tools_4.4.1        generics_0.1.3
##  [9] parallel_4.4.1     proxy_0.4-27       pkgconfig_2.0.3    Matrix_1.7-0
## [13] KernSmooth_2.23-24 RColorBrewer_1.1-3 S7_0.2.0           lifecycle_1.0.4
```

```
## [17] compiler_4.4.1     farver_2.1.2       textshaping_0.4.1  tinytex_0.54
## [21] htmltools_0.5.8.1  class_7.3-22       yaml_2.3.10        crayon_1.5.3
## [25] pillar_1.11.0      classInt_0.4-11    gtools_3.9.5       tidyselect_1.2.1
## [29] digest_0.6.37      stringi_1.8.4      labeling_0.4.3     fastmap_1.2.0
## [33] grid_4.4.1         cli_3.6.5          magrittr_2.0.3     withr_3.0.2
## [37] scales_1.4.0       bit64_4.6.0-1      timechange_0.3.0   rmarkdown_2.29
## [41] bit_4.5.0.1        hms_1.1.3          evaluate_1.0.3     viridisLite_0.4.2
## [45] rlang_1.1.6        Rcpp_1.1.0         glue_1.8.0         DBI_1.2.3
## [49] xml2_1.3.6         vroom_1.6.5        svglite_2.2.1      rstudioapi_0.17.1
## [53] R6_2.6.1           plyr_1.8.9         systemfonts_1.3.1  units_0.8-7
```